

From Awareness to Conversion: How Firm Characteristics Shape Advertising Budget Allocation
Across the Marketing Funnel

By
Othmane Oumnad

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Advised By
Pr. Colin Davison

Department of Economics
The College of Wooster

Research Question

How do firm characteristics shape the allocation of advertising budgets across the marketing funnel stages?

Theory Chapter

Introduction

Effective allocation of marketing resources across the different stages of the customer journey is a central challenge for firms in today's competitive and multi-channel environment. While investing in advertising can help move potential customers from initial awareness to long-term retention, knowing how much to spend at each stage, and understanding what determines this balance, is a complex problem with substantial impact on overall profitability.

This chapter tackles the question of how firm characteristics such as size, age, and industry influence the optimal distribution of advertising budgets across the main funnel stages: awareness, consideration, conversion, and retention. Specifically, it develops a flexible economic model that allows these characteristics to shape both the allocation strategy and the effectiveness of marketing investments at every step.

The theoretical journey begins with the foundational Dorfman-Steiner model, which sets the classic benchmark for resource allocation in marketing. Building on this, the chapter introduces a stage-by-stage framework using logit response functions, which capture how advertising spend impacts progression probabilities in a realistic, non-linear way. These probabilities, and their parameters, reflect concrete firm-level differences, making the model adaptable to a variety of organizational contexts.

To guide both intuition and practical implementation, the analysis first examines a simplified two-stage case without a budget constraint, highlighting the key trade-offs in allocation. It then

generalizes to the full multi-stage model with an explicit budget constraint to reflect the realities firms face when resources are limited.

This theory chapter lays the groundwork for the empirical and policy analysis that follow in later chapters of the independent study. By clarifying the link between firm characteristics, marketing structure, and allocation outcomes, it provides a framework for data-driven decision making and deeper academic understanding of modern marketing strategy.

Dorfman–Steiner Model: Mathematical Derivation

Effective marketing always faces a fundamental question: How much should a firm invest in advertising, and how should that decision interact with the price it sets for its products? As markets became more complex and competition grew, economists needed a rigorous, quantitative framework to describe the relationship between advertising, pricing, and profit.

The Dorfman-Steiner (DS) model, developed in the mid-20th century, was one of the first to give a mathematical answer to this question. Its impact has been lasting, serving as a foundational tool in both economics and marketing science. What makes it especially valuable is its ability to reduce an ambiguous managerial challenge into a precise allocation rule based on observable properties of demand. The DS model shows how profit-maximizing advertising and pricing decisions are systematically linked through the sensitivity of demand to both advertising effort and price changes.

Within this model, advertising is not treated as a separate afterthought, but as a variable that directly interacts with pricing, shaping demand and total profit. This integrated perspective provides more reliable and actionable guidance than intuition or tradition. Managers, according to the DS framework, should scale their advertising expenditures in direct proportion to how responsive their customers are to advertising, and relative to how sensitive they are to price.

In the context of this independent study, the D-S model serves as a clear starting point for analyzing optimal resource allocation. By formalizing the link between elasticities (for price and advertising) and optimal budgets, it offers a transparent benchmark from which more sophisticated, multi-stage allocation frameworks can be developed. Understanding this baseline

model is crucial before expanding to the multi-stage, funnel-based settings that better reflect modern marketing environments.

Assume a profit-maximizing firm operating in a monopolistically competitive market that sells its product at price p with constant marginal cost c . Total advertising expenditure is A , and demand (quantity sold) q is a function of both price and advertising, $q = q(p, A)$, with $q_p < 0$ and $q_A > 0$. The cost of advertising is linear in A , and the firm chooses p and A simultaneously to maximize profit.

The profit function is:

$$\pi = (p - c) \cdot q(p, A) - A$$

To find the optimal price, we first compute the first-order condition with respect to p :

$$\frac{\partial \pi}{\partial p} = q(p, A) + (p - c) \frac{\partial q}{\partial p}$$

Setting this derivative equal to zero for maximization:

$$q(p, A) + (p - c)q_p = 0$$

Rearranging and dividing both sides by pq , we express the profit margin as:

$$\frac{p - c}{p} = - \frac{q(p, A)}{p \cdot \frac{\partial q}{\partial p}}$$

Recognize that $\epsilon_p = \frac{\partial q}{\partial p} \cdot \frac{p}{q}$ is the price elasticity of demand (negative):

$$\frac{p - c}{p} = - \frac{1}{\epsilon_p}$$

This result aligns the optimal markup with the inverse of the price elasticity, a classic foundation in microeconomic pricing theory.

Next, we compute the first-order condition with respect to advertising expenditure A

$$\frac{\partial \pi}{\partial A} = (p - c) \frac{\partial q}{\partial A} - 1$$

Set to zero for maximization:

$$(p - c) \frac{\partial q}{\partial A} = 1$$

dividing both sides by pq to normalize, we write:

$$\frac{p - c}{p} \cdot \frac{\frac{\partial q}{\partial A}}{q} = \frac{1}{pq}$$

Recognizing that $\epsilon_A = \frac{\partial q}{\partial A} \cdot \frac{A}{q}$ is the advertising elasticity of demand, we express the marginal effect of advertising as:

$$\frac{\partial q}{\partial A} = \epsilon_A \cdot \frac{q}{A}$$

Substitute into the first-order condition:

$$(p - c) \cdot \epsilon_A \cdot \frac{q}{A} = 1$$

Rearrange for the advertising-to-sales ratio:

$$\frac{A}{pq} = \frac{p - c}{p} \cdot \epsilon_A$$

The core result of the Dorfman-Steiner condition dictates that a firm's advertising-to-sales ratio should be proportional to the profit margin and advertising elasticity. Put simply, firms should invest more in advertising when their product margins and advertising effectiveness are high, and less when demand is more price sensitive. This theoretical benchmark helps managers balance spending between price promotion and advertising investment to maximize profit.

We can also express the optimal advertising-to-sales ratio in an intuitive way, using the language of elasticities. Substituting the profit margin condition ($\frac{p-c}{p} = -\frac{1}{\epsilon_p}$) into the Dorfman-Steiner equation, we obtain:

$$\frac{A}{pq} = \frac{\epsilon_A}{\epsilon_p}$$

This expression shows that the optimal proportion of sales revenue devoted to advertising equals the ratio of the advertising elasticity of demand to the (negative) price elasticity of demand. In practical terms, when customers are highly responsive to advertising relative to their sensitivity to price changes, the firm should allocate a larger share of its revenue to advertising. This compact ratio provides a powerful, actionable rule connecting two fundamental market responses to real managerial decisions.

While profoundly influential, the classic Dorfman-Steiner model considers only the aggregate effect of advertising across an undifferentiated market and a single decision stage. Contemporary marketing challenges, especially in digital channels, require finer allocation across multiple funnel stages and attention to firm-specific factors that affect advertising effectiveness. In the following sections, I extend this analysis to a multi-stage framework, modeling each stage's effectiveness with response functions whose parameters reflect underlying firm characteristics. This provides richer insight into how firms should adapt resource allocation for maximum profit in complex, real-world customer journeys.

Stage-Specific Advertising Effectiveness: The Logit Response Model

The marketing funnel is a conceptual model that represents the customer journey from initial awareness of a product or brand through consideration, conversion, and ultimately retention. Each stage poses its own challenges and opportunities, and firms can influence the movement of potential customers across these stages through targeted advertising and promotional efforts. Modern marketing strategy increasingly recognizes the importance of optimizing spend at each funnel stage, rather than treating advertising as a single aggregate investment.

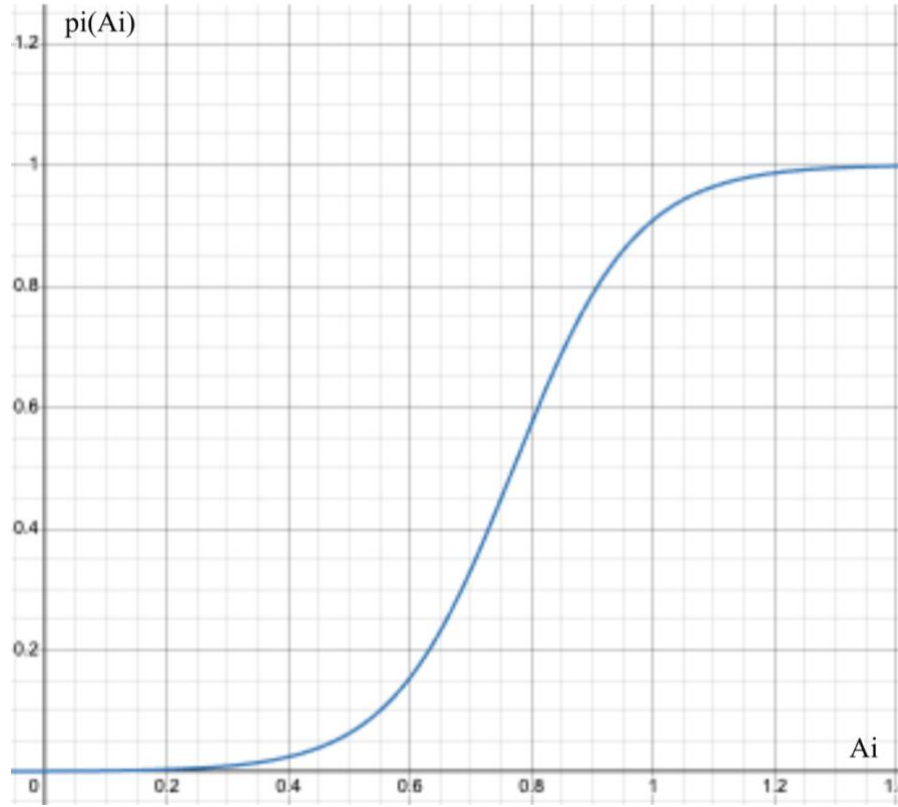
Advertising budget allocation within the marketing funnel must be understood as a dynamic process in which each stage operates both independently and as part of a larger system. At every stage, awareness, consideration, conversion, and retention, a firm's decision about how much to invest in advertising does not simply drive direct outcomes, but also shapes the pool of customers available for progression in all following stages.

In this framework, each stage is modeled as a probabilistic filter. The probability that a customer will advance to the next level is not static but is directly influenced by the financial resources targeted at that stage. This approach captures real-world uncertainty: not every individual exposed to marketing will respond uniformly, and the effectiveness of spending depends both on the inherent tendencies of the market and the firm's own choices. By embedding this uncertainty within the model, the analysis acknowledges the variability and complexity typical of consumer behavior.

The logit response function is adopted both for its analytic tractability and its ability to mimic empirically observed patterns in advertising effectiveness. Specifically, the logit is S-shaped, which means that at very low or very high spending levels, increases in investment deliver only modest improvements in progression rates. It is only around an intermediate, critical point, often corresponding to optimal campaign design or sufficient market coverage, that marginal returns to spending peak, and additional funds have the greatest impact. This property makes the logit specification especially relevant in economic contexts where threshold effects and diminishing returns are present.

More formally, for each stage i , the probability of customer advancement is given by:

$$p_i(A_i) = \frac{1}{1 + e^{-(a_i + b_i A_i)}}$$

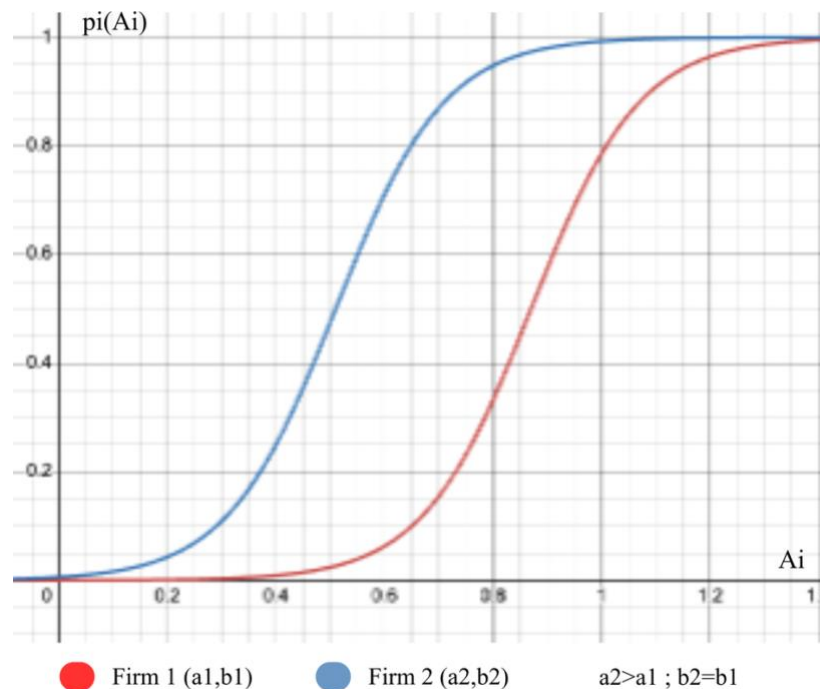


Graph 1: Stage-Specific Advertising Effectiveness

The parameters a_i and b_i in this specification are fundamentally shaped by a firm's unique characteristics, and play a critical role in determining the stage-level effectiveness of advertising. In this framework, firm characteristics are interpreted broadly. They can include observable attributes such as firm size, age, industry sector, brand reputation, and historical marketing effectiveness, as well as less tangible features like organizational agility, digital maturity, or customer relationship management capabilities. Specifically, a_i reflects how intrinsically challenging it is for a given firm to move customers through stage i in the absence of advertising, while b_i encodes how responsive progression at that stage is to incremental investments. By viewing these parameters as functions of underlying firm characteristics, the model naturally accommodates cross-firm differences in funnel response and enables the analysis of how diverse organizational factors translate into measurable budget allocation effects.

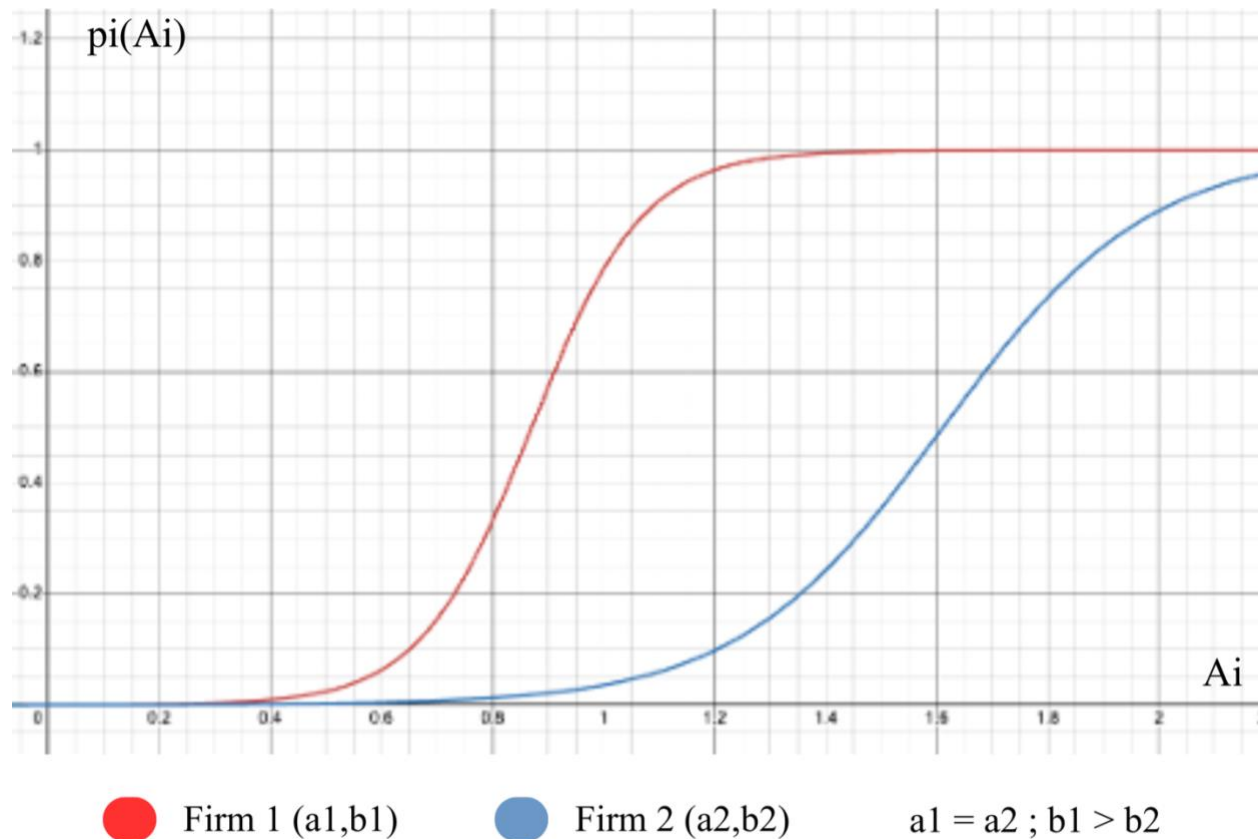
To illustrate how firm characteristics map into a and b , consider two simple pairwise comparisons of stage- k logit schedules.

First, hold responsiveness fixed at $b_1 = b_2$ and compare firms that differ only in baseline difficulty, with $a_2 > a_1$. The red curve representing firm 1 (a_1, b_1) and the blue curve firm 2 (a_2, b_2). The blue curve lies uniformly above the red curve, indicating that, for any given advertising outlay at stage k , the higher- a firm converts a larger share of customers. This configuration corresponds to an “established” firm that faces fewer frictions in moving customers through the stage even without advertising, for example because of strong brand equity or an existing installed base.



Second, hold baseline difficulty fixed at $a_1 = a_2$ and compare firms that differ only in responsiveness, with $b_1 > b_2$. Here, again with the red graph for firm 1 and blue graph for firm 2. Now the curves cross in slope rather than in level: both firms start from the same baseline probability at zero spend, but the red curve steepens more quickly, implying that marginal returns to additional advertising are higher for firm 1 at intermediate spending levels. This

pattern is consistent with a more agile or digitally mature organization that is able to translate incremental budget into effective impressions, even if its underlying baseline position at stage k is no better than that of its rival.



Quantifying Funnel Progression: Probabilistic Funnel Modeling

In the marketing funnel framework, the customer journey is conceptualized as a sequence of stages, each representing a hurdle that potential customers must clear to become loyal buyers. At the beginning, a firm targets a broad audience (N_0), but not all individuals will advance through each stage. Instead, at every point in the funnel, only a fraction of the original pool moves forward, determined by the probability of progression for that stage. These progression probabilities, $p_i(A_i)$, depend on the advertising investment at stage i and on how effective that investment is, which in turn reflects both market factors and firm-specific characteristics.

The probabilistic nature of the funnel means that the overall chance of retaining a customer is the product of the probabilities of passing each stage. As a result, the expected outcome at the end of the funnel, such as the number of retained or converted customers, is calculated by multiplying the initial audience size by the progression probabilities at all successive stages. This structure highlights how changes or improvements at any single stage can have a compounding effect on overall performance, while a weak stage can substantially reduce the total outcome.

Mathematically, the expected number of retained customers is:

$$N_4 = N_0 \cdot p_1(A_1) \cdot p_2(A_2) \cdot p_3(A_3) \cdot p_4(A_4)$$

where N_0 is the initial target audience and each $p_i(A_i)$ is the progression probability for stage i .



The multiplicative structure of the funnel model serves not simply as a mathematical convenience, but as a clear reflection of stagewise interdependence. Improvements (or weaknesses) at any single stage propagate through the entire funnel: a strong initial investment increases the pool of prospects for future conversion, while a neglected stage can bottleneck overall performance. As such, optimal allocation cannot be achieved by considering stages in isolation but must be approached as a coordinated system.

Optimization Model: Two-Stage Advertising Allocation

Having established how advertising spend at each funnel stage produces measurable outcomes, we now turn to the central question of how firms can optimally allocate resources to maximize profit. This transition from modeling outcomes to guiding decision-making lies at the heart of managerial analytics, as it provides actionable rules for budget distribution and thus strategic impact. Recognizing both the mathematical and practical complexity of multi-stage funnels, this section focuses on a simplified optimization model with only two critical stages: the upper funnel (awareness) and the lower funnel (conversion). This reduction serves two purposes. First, it makes the optimization problem analytically tractable, revealing the logic and intuition that would otherwise be obscured in a more general setting. Second, by temporarily omitting the budget constraint, we isolate the influence and interaction of stage-level investment decisions without confounding factors, thereby clarifying how firm characteristics and marketing effectiveness shape the optimal strategy. The resulting framework provides a foundation for understanding more complex, realistic allocation problems, which will be addressed in subsequent sections.

The central objective for the firm is to choose budget levels A_1 and A_2 , allocated to awareness and conversion respectively, so as to maximize expected profit. Formally, the problem is:

$$\max_{A_1, A_2} \Pi = P \cdot N_0 \cdot p_1(A_1) \cdot p_2(A_2) - (A_1 + A_2)$$

Where:

- P : profit per converted customer
- N_0 : initial number of customers entering the funnel
- $p_1(A_1)$: probability of moving to the awareness stage,
- $p_2(A_2)$: probability of moving to the conversion stage,

The logit functional forms for stage response probabilities are:

$$p_1(A_1) = \frac{1}{1 + e^{-(a_1 + b_1 A_1)}}$$

$$p_2(A_2) = \frac{1}{1 + e^{-(a_2 + b_2 A_2)}}$$

To find the optimal budget allocation, we set the first-order conditions (FOCs):

$$\frac{\partial \Pi}{\partial A_1} = P N_0 \frac{\partial p_1(A_1)}{\partial A_1} p_2(A_2) - 1 = 0$$

$$\frac{\partial \Pi}{\partial A_2} = P N_0 p_1(A_1) \frac{\partial p_2(A_2)}{\partial A_2} - 1 = 0$$

The derivative of the logit probability function is:

$$\frac{\partial p_i(A_i)}{\partial A_i} = \frac{b_i e^{-(a_i + b_i A_i)}}{(1 + e^{-(a_i + b_i A_i)})^2}$$

Setting the first-order conditions equal gives:

$$P \cdot N_0 \cdot \frac{\partial p_1(A_1)}{\partial A_1} \cdot p_2(A_2) - 1 = P \cdot N_0 \cdot p_1(A_1) \cdot \frac{\partial p_2(A_2)}{\partial A_2} - 1$$

This simplifies to:

$$\frac{\partial p_1(A_1)}{\partial A_1} \cdot p_2(A_2) = p_1(A_1) \cdot \frac{\partial p_2(A_2)}{\partial A_2}$$

Replacing the derivatives with their logit expressions gives:

$$\frac{b_1 e^{-(a_1 + b_1 A_1)}}{(1 + e^{-(a_1 + b_1 A_1)})^2} \cdot \frac{1}{1 + e^{-(a_2 + b_2 A_2)}} = \frac{1}{1 + e^{-(a_1 + b_1 A_1)}} \cdot \frac{b_2 e^{-(a_2 + b_2 A_2)}}{(1 + e^{-(a_2 + b_2 A_2)})^2}$$

Thus, the optimality condition for the joint allocation is:

$$\frac{b_1 e^{-(a_1 + b_1 A_1)}}{(1 + e^{-(a_1 + b_1 A_1)})} = \frac{b_2 e^{-(a_2 + b_2 A_2)}}{(1 + e^{-(a_2 + b_2 A_2)})}$$

The joint optimality condition derived in this two-stage setting provides clear managerial insight: at the optimal allocation, the marginal profit generated by an additional unit of advertising spend must be equalized across both the awareness and conversion stages. This ensures that no resources are left unexploited in one stage while yielding higher incremental returns in the other. Importantly, the parameters a_1, a_2, b_1 , and b_2 , which capture observable firm characteristics such as company age, size, and industry sector, directly influence both the allocation rule and its effectiveness. For example, larger or more established firms, or those operating in dynamic sectors, may experience different baseline awareness or responsiveness to spend at different funnel stages, which justifies varying investment strategies. Despite its analytical clarity, this model's main limitation is its abstraction from real-world budget constraints and from more granular, multi-stage dynamics. As such, it serves as a theoretical baseline, offering intuition and structure but necessitating further extension to handle the full complexity of practical multi-channel marketing allocation.

To provide intuition for the optimality condition, we analyze how changes in the model parameters a_1, a_2, b_1, b_2 influence the relationship between the optimal budgets allocated to awareness (A_1) and conversion (A_2). The key condition governing optimal joint allocation is:

$$\frac{b_1 e^{-(a_1 + b_1 A_1)}}{(1 + e^{-(a_1 + b_1 A_1)})^2} = \frac{b_2 e^{-(a_2 + b_2 A_2)}}{(1 + e^{-(a_2 + b_2 A_2)})^2}$$

We now illustrate how the parameters affect the solution:

Case 1 (full symmetry): $a_1 = a_2$ and $b_1 = b_2$.

In this case the two logit functions are exactly the same, so $p_1(A) = p_2(A)$ and $\partial p_1(A)/\partial A = \partial p_2(A)/\partial A$ for all spending levels A . The profit function and the optimality conditions are therefore invariant to swapping the labels “1” and “2”, which implies that if (A_1^*, A_2^*) is an interior optimum then (A_2^*, A_1^*) is also an interior optimum; strict concavity then forces these two to coincide, so the unique interior solution satisfies

$$A_1^* = A_2^*.$$

Economically, when awareness and conversion respond identically to spending and each additional dollar of budget has the same diminishing-returns pattern in both stages, the only way to equalize marginal profit is to spend equally in the two stages.

Case 2 (awareness is more responsive): $a_1 = a_2$ but $b_1 > b_2$.

Holding intercepts fixed, a higher slope parameter b_1 makes the awareness probability $p_1(A_1)$ increase faster with spending than the conversion probability $p_2(A_2)$ increases with A_2 . For any common spending level A , this implies both a larger marginal increase in awareness probability $\partial p_1(A)/\partial A$ and a higher level of awareness probability $p_1(A)$ relative to conversion, so the marginal profit from a dollar in stage 1 would strictly exceed that in stage 2 if $A_1 = A_2$. To restore equality of marginal profit across stages, the optimal allocation must tilt spending away from the conversion stage and toward the awareness stage, in order to increase the marginal profit of stage 2 and decrease the marginal profit of stage 1 until they equalize

$$A_1^* > A_2^*.$$

In economic terms, when upper-funnel advertising is “stronger” (each dollar generates a larger incremental change in awareness), the firm needs to spend more there to reach the point where additional dollars are no more profitable than additional dollars in conversion, so the optimal plan assigns a strictly smaller budget to conversion than to awareness.

Case 3 (awareness has higher baseline): $a_1 > a_2$ and $b_1 = b_2$.

With equal slopes, changing the intercepts shifts the two logit curves vertically while preserving their curvature and the rate at which marginal returns decline with spending. Because the logit probability is increasing in the intercept, $a_1 > a_2$ means that for any given spending level A , the awareness probability $p_1(A)$ and its marginal effect $\partial p_1(A)/\partial A$ are both strictly higher than the corresponding conversion values $p_2(A)$ and $\partial p_2(A)/\partial A$.

The marginal-profit equality is

$$\frac{be^{-(a_1+bA_1)}}{1+e^{-(a_1+bA_1)}} = \frac{be^{-(a_2+bA_2)}}{1+e^{-(a_2+bA_2)}}.$$

Key simplification

Cancel $b \neq 0$ on both sides and define $x_i = a_i + bA_i$.

The equality becomes

$$\frac{e^{-x_1}}{1 + e^{-x_1}} = \frac{e^{-x_2}}{1 + e^{-x_2}}.$$

The function $f(x) = \frac{e^{-x}}{1+e^{-x}}$ is strictly monotone in x , so the only way this equality can hold is if

$$x_1 = x_2 \Rightarrow a_1 + bA_1 = a_2 + bA_2.$$

Implication for A_1 vs A_2

Rearrange:

$$b(A_1 - A_2) = a_2 - a_1.$$

- If $b > 0$ (higher A increases the stage outcome), then $a_1 > a_2 \Rightarrow a_2 - a_1 < 0$, so

$$A_1 - A_2 < 0 \Rightarrow A_1^* < A_2^*.$$

- If $b < 0$, the inequality reverses and $A_1^* > A_2^*$.

So, under the usual assumption that advertising has a positive marginal effect in both stages ($b > 0$), a larger intercept a_1 must be offset by a *smaller* optimal spend A_1 than A_2 to restore equality of marginal profits

Full Funnel Optimization: The General Model with Budget Constraint

The previous sections illustrated how optimal advertising allocation can be determined in a simple two-stage funnel. While this approach helps build intuition, real marketing funnels typically consist of several stages, each with its own role and challenges. Here, we extend the model to its general form, incorporating all four key stages; Awareness, Consideration, Conversion, and Retention. Including a budget constraint to reflect practical resource limits.

In this setting, the firm's goal is to decide how much of its total available budget to allocate to each stage in order to maximize overall profit. Let A_1, A_2, A_3, A_4 represent the advertising spend at each funnel stage. The optimization problem becomes:

$$\max_{A_1, A_2, A_3, A_4} \Pi = P \cdot N_0 \cdot p_1(A_1) \cdot p_2(A_2) \cdot p_3(A_3) \cdot p_4(A_4) - (A_1 + A_2 + A_3 + A_4)$$

subject to:

$$A_1 + A_2 + A_3 + A_4 \leq B$$

To identify the optimal allocation across all four stages, we apply the method of Lagrange multipliers, introducing a multiplier λ that enforces adherence to the budget:

$$\mathcal{L}(A_1, A_2, A_3, A_4, \lambda) = P \cdot N_0 \prod_{i=1}^4 p_i(A_i) - (A_1 + A_2 + A_3 + A_4) + \lambda[B - (A_1 + A_2 + A_3 + A_4)]$$

Solving this model tells the firm exactly how to allocate limited resources across every step of the funnel to maximize its expected profit. In theory, the optimal allocation is where the marginal profit per dollar spent is balanced across all stages, accounting for how spend at each stage boosts the probability of final retention.

However, solving for the optimal values of A_1, A_2, A_3 , and A_4 cannot be done in closed form because the product of nonlinear logit functions makes the math too complex. Instead, the firm must use numerical methods, such as computational algorithms, to find the solution. Using actual values for a_i and b_i that capture its own size, age, industry, and market environment.

This generalized model delivers a realistic and adaptable framework for marketing allocation. It demonstrates not only the theoretical logic but also the practical steps required to apply advanced models in real business contexts.

Conclusion

This theory chapter developed a comprehensive framework for understanding optimal marketing budget allocation across the stages of the customer journey. Beginning with the foundational

Dorfman-Steiner model and extending to a multi-stage structure with logit-based response functions, the analysis highlighted how firm characteristics such as size, age, and industry, play a decisive role in shaping both allocation strategies and the effectiveness of marketing investments at each stage.

A central insight is that optimal budget allocation requires equalizing the marginal profit from advertising spending across funnel stages, always accounting for the unique attributes of each firm. By treating the customer journey as a sequence of progression probabilities influenced by both managerial choice and underlying firm factors, the models offer clear, actionable principles for resource distribution in complex market environments.

Nonetheless, several limitations deserve attention. The models here rely on simplifying options, including homogeneous audiences, independent funnel stages, and specific functional forms for advertising response. These abstractions, while useful for generating tractable results, may not fully capture the richer realities of multi-channel marketing, where consumer behavior is more diverse and interactions across channels or over time can be complex.

Despite these boundaries, the theoretical structure presented here provides a solid starting point for both further research and practical application. It brings conceptual clarity to the allocation problem and invites both empirical testing and future model extensions. Ultimately, these contributions help bridge economic theory and managerial decision making in the evolving domain of marketing strategy.

Literature Review

Article 1: Yurtseven and Günalp (2023) – Determinants of advertising intensity in Turkish manufacturing

1. Introduction

This article review examines Yurtseven and Gunalp's (2023) study, "A panel data analysis of the determinants of advertising intensity in the Turkish manufacturing sector," in relation to this IS on optimal marketing budget allocation across the customer journey. Drawing on the Dorfman–Steiner condition, which links a firm's optimal advertising-to-sales ratio to the product of its price–cost margin and the advertising elasticity of demand, this review uses the paper as an empirical test of how that rule operates in an emerging market setting with industry-level panel data.

The article offers rare, rigorous evidence on the Dorfman–Steiner condition outside advanced economies, establishing a clear statistical connection between profitability, market structure, and advertising intensity. This connection provides the empirical backbone for this IS by validating the assumption that budget allocation across marketing activities is constrained by basic economic variables such as price–cost margins and competition.

Yurtseven and Gunalp find a positive, statistically significant relationship between price–cost margins and advertising intensity, supporting the Dorfman–Steiner prediction that more profitable firms advertise more. They also document an inverted-U relationship between market concentration and advertising intensity, indicating that competition shapes advertising investment in a nonlinear way rather than through a simple monotonic effect, which motivates the inclusion of firm heterogeneity, nonlinearity, and industry structure in the multi-stage and multi-channel allocation framework developed in this IS.

2. Detailed Critical Description and Analysis

2.1 Article Overview and Objectives

Yurtseven and Gunalp (2023) investigate the determinants of advertising intensity in the Turkish manufacturing sector. Their primary objective is to test the Dorfman–Steiner condition empirically and to examine the relationship between advertising intensity and market concentration. The article situates itself within the classical literature on optimal advertising allocation, extending the analysis to an emerging market context with industry-level panel data.

2.2 Methods and key equations

The authors utilize panel data covering 66 four-digit manufacturing industries in Turkey from 1993 to 1999. The dataset is constructed from industry-level sources, including advertising expenditures, total output, price–cost margins, concentration ratios, growth rates, and indicators for consumer and durable goods. Additional controls include R&D intensity and export and import shares. The temporal and industry breadth of the dataset allows for examination of both cross-sectional and longitudinal variations in advertising behavior.

The primary econometric model employed is a fixed effects linear regression of advertising intensity on several key variables. The specification is:

$$\text{Advertising Intensity} = \alpha + \beta_1 \text{ Price-Cost Margin} + \beta_2 \text{ Concentration Ratio} + \beta_3 (\text{Concentration Ratio})^2 + \beta_4 \text{ Growth Rate} + \beta_5 \text{ Consumer Dummy} + \beta_6 \text{ Durable Dummy} + \beta_7 \text{ R\&D Intensity} + \beta_8 \text{ Import Intensity} + \beta_9 \text{ Export Intensity} + u$$

This model is estimated with fixed effects to control for unobserved industry heterogeneity.

2.3 Results and Interpretation

In their core fixed effects specification (Table 3), the authors obtain the following estimated coefficients:

- Price–cost margin: 0.0412 ($p < 0.05$)

- Concentration ratio: 0.0997 ($p < 0.05$)
- Concentration ratio squared: -0.0067 ($p < 0.10$)
- Growth rate: 0.0027 ($p < 0.05$)
- Export intensity: -0.0110 ($p < 0.10$)

Other variables, such as R&D intensity and the dummy variables, do not reach statistical significance in their main model.

The positive and significant coefficient for price–cost margin confirms the Dorfman–Steiner condition: as profitability rises, firms increase their advertising expenditures relative to sales. The quadratic form of the concentration ratio reveals an inverted-U relationship, meaning advertising intensity is highest in moderately concentrated industries and declines as competition falls or increases beyond a threshold. The negative effect of export intensity suggests that industries facing international competitive pressures may allocate less expenditure to advertising domestically. These findings are broadly consistent with classical theory but also reveal context-specific nuances relevant to developing economies.

2.4 Flaws and Limitations

A key limitation is the use of industry-level rather than firm-level data. This approach conceals within-industry variation in advertising strategy and may introduce aggregation bias. The lack of direct measures for advertising elasticity requires reliance on the price–cost margin as a proxy, which may weaken the interpretation. Additionally, only traditional media advertising is covered, excluding rapidly expanding digital channels.

The study examines Turkish manufacturing industries in the 1990s, a period and context with specific regulatory, technological, and competitive dynamics. The results may not fully generalize to other sectors, countries, or periods, particularly in more advanced economies or digital-dominated markets. Thus, while the findings support the classical condition, the magnitude and structure of effects should be interpreted cautiously.

2.5 Contributions and strengths

The article's use of panel data and fixed effects robustly controls for unobserved heterogeneity among industries, strengthening the validity of its results. The explicit testing of both the Dorfman–Steiner condition and the inverted-U hypothesis provides value by rigorously evaluating theoretical predictions. The careful attention to model specification and sensitivity checks adds credibility.

Yurtseven and Gunalp contribute to a limited but important body of empirical work on advertising allocation in emerging markets. Their findings closely align with established results from advanced economies, suggesting that key economic mechanisms are robust across contexts. At the same time, their results offer new insights into the effect magnitude and competitive structure in Turkish manufacturing, which enriches comparative analysis.

2.6 Relevance

The confirmation of Dorfman–Steiner's central relationship provides an empirical foundation for my research model. The observed nonlinear effects of competition justify the incorporation of nonlinearity in modeling budget allocation across multiple channels and funnel stages.

This empirical work encourages the explicit inclusion of profitability and market structure variables in my own regression models. It also motivates a broader search for data that can allow firm-level, channel-specific, or funnel-stage allocation analysis. The observed shortcomings affirm the need for extensions that better capture heterogeneity and the role of digital marketing, informing both hypothesis refinement and empirical strategy.

3. Conclusion

This article provides clear empirical support for the Dorfman–Steiner condition, demonstrating that advertising intensity in Turkish manufacturing industries increases with price–cost margin and displays a nonlinear relationship with market concentration. The findings confirm that economic theory remains relevant in diverse market contexts, though the size and specifics of

effects may vary. The limitations arising from industry-level data and context-specific details are evident, underscoring the importance of firm-level analysis and the inclusion of new marketing channels. Nonetheless, the study's rigorous panel methodology and detailed exploration of classical advertising theory offer useful benchmarks for understanding advertising allocation and its determinants.

Yurtseven and Gunalp's empirical test of advertising intensity determinants provides a valuable anchor for the theoretical and empirical development of my independent study. The article strengthens the foundation of my modeling approach by confirming that profitability and market structure are quantitatively important drivers of budget allocation. It also motivates extensions that better capture heterogeneity and the dynamics of newer marketing environments. The study's strengths in estimation and theoretical grounding outweigh its contextual limitations, making it a substantive contribution to the literature review and an informative reference point for my research design and analysis.

Article 2: Abhishek, Fader, and Hosanagar – Media exposure through the funnel and multi-stage attribution

1. Introduction

Attribution in digital marketing has become a central topic in both economics and quantitative marketing. Firms allocate large budgets across heterogeneous media channels while tracking consumers through multi-stage purchase funnels. Yet most attribution models treat these funnel stages as either independent or non-causal, and many do not account for the fact that consumer progression is sequential, path-dependent, and shaped by distinct types of media exposures at each stage. *Media Exposure Through the Funnel: A Model of Multi-Stage Attribution* addresses this gap by proposing an econometric framework in which advertising influences consumers differently as they transition through a multi-stage funnel. The paper develops a structural model in which exposures to search, display, and video ads affect three distinct consumer states: no-

visit, site-visit, and conversion. Consumers evolve through these states according to a Markov process whose transition probabilities are influenced by accumulated ad impressions.

The contribution of the article lies in its explicit modeling of the funnel structure. The authors argue that advertising affects consumers differently depending on their current stage, and that correct attribution must consider how exposures change the probability of movement through the sequential funnel. This stands in contrast to last-click attribution, which assigns full credit to the final interaction, and to aggregate models that estimate average incremental effects without accounting for funnel dynamics. By embedding ad exposures in structural transition probabilities, the authors attempt to recover causal effects under specific assumptions and generate stage-specific marginal returns on advertising. These returns then inform optimal budget allocation, which connects directly to the allocation questions in this IS.

2. Detailed Critical Description and Analysis of the Article

2.1 Article Overview and Objectives

Abhishek et al. examine how repeated online advertising exposures shape a consumer's progression toward conversion and how to assign credit to these exposures. The paper positions the attribution problem as one of measuring incremental impact rather than relying on rule-of-thumb methods such as last-touch attribution. The authors adopt a multi-stage funnel as the behavioral structure guiding consumer movement. In this framework, advertising affects the probability of transitioning between latent stages of engagement, and the overall likelihood of eventual conversion depends on both these transitions and state-conditional behaviors.

The central objectives are: (i) to estimate how different ad formats (display, search, video) influence transitions between funnel stages and state-conditional conversion probabilities, and (ii) to derive a model-based attribution rule that assigns incremental value to each ad exposure along a user's path. The ultimate goal is to show how attribution profiles change once funnel dynamics and heterogeneity are explicitly modeled, and to assess the implications for channel evaluation and budget allocation.

2.2 Methods and key equations

The empirical analysis is based on an eleven-week online campaign for the launch of a new car model. The campaign features display advertising on both general-interest websites (Yahoo, MSN, Facebook) and automotive-specific websites (Kelley Blue Book, Edmunds), along with paid search ads. User-level data link all impressions, display clicks, search clicks, website visits, and conversion events through cookies, yielding detailed histories of exposure and behavior for each individual. The sample includes 6,432 randomly selected users and 146,165 ad or website-related events.

The funnel is modeled as a Hidden Markov Model (HMM), in which consumers occupy unobserved latent states that evolve over time. The states correspond to increasing levels of engagement: disengaged, active, engaged, and converted, with conversion treated as an absorbing state. Let S_{it} denote the latent state of user i at time t , and Y_{it} the observed outcomes, consisting of page views and conversion indicators.

The latent state follows a first-order Markov process with time-varying transition probabilities. For user i , the transition from state j to state k at time t is modeled as a multinomial logit in advertising covariates A_{it} :

$$P(S_{i,t+1} = k \mid S_{it} = j, A_{it}) = \frac{\exp(\alpha_{jk} + A'_{it}\beta_{jk} + \mu_{ik})}{\sum_{m \in J(j)} \exp(\alpha_{jm} + A'_{it}\beta_{jm} + \mu_{im})},$$

where $J(j)$ denotes the set of states reachable from j , β_{jk} captures the effect of each ad type on the transition, and μ_{ik} is an individual-specific intercept absorbing unobserved heterogeneity.

Page views are modeled as Poisson with a state- and ad-dependent rate:

$$Y_{it}^{\text{page}} \mid S_{it} = s \sim \text{Poisson}(\lambda_{is}), \ln(\lambda_{is}) = \gamma_s + A'_{it}\theta_s + \eta_{r(i)},$$

where γ_s is a baseline research intensity for latent state s , θ_s measures how ad exposures influence this intensity, and $\eta_{r(i)}$ is a region-level random effect. Parameters are constrained so that page-view intensity rises as consumers move through the funnel.

Conversion is modeled as state-dependent and binary:

$$P(Y_{it}^{\text{conv}} = 1 \mid S_{it} = s) = \frac{\exp(\delta_s + X'_{it}\kappa_s + \xi_{d(i)})}{1 + \exp(\delta_s + X'_{it}\kappa_s + \xi_{d(i)})},$$

where X_{it} includes advertising covariates and page views, and $\xi_{d(i)}$ is a DMA-level random effect. Conversion probabilities are constrained to increase with the latent state index, reinforcing the funnel interpretation. Together, these components create a dynamic system in which advertising influences conversion both directly (through the conversion equation) and indirectly (by shifting the distribution of consumers across latent states).

2.3 Results and interpretation

The estimated transition model reveals distinct roles for different ad formats. Display impressions, especially on automotive websites, meaningfully increase the likelihood that disengaged consumers transition to the active state. These effects occur even without subsequent website visits, suggesting that display ads generate early engagement and awareness. This supports the common view of display as an upper-funnel driver.

Search clicks and display clicks have larger effects later in the funnel. Search clicks substantially raise the probability of moving from active to engaged and also increase state-conditional conversion rates. For example, when a consumer clicks a search ad, the likelihood of moving from active to engaged rises from roughly 0.10 to about 0.14, and conversion probabilities among engaged consumers increase even more sharply. The results support an interpretation of search and clicks as mid- and lower-funnel instruments that help close the sale once interest has been formed.

The HMM structure allows the authors to compute the incremental value of each ad event. For ad a experienced by user i at time t , its value is defined as:

$$v_{iat} = P(\text{convert} \mid \text{ad present}) - P(\text{convert} \mid \text{ad removed}),$$

where both probabilities are evaluated using the estimated model and integrate over potential future sequences of events. Aggregating these values across users and channels yields a model-based attribution profile. Relative to last-touch attribution, search clicks receive much less credit, while early-funnel display impressions and display clicks receive substantially more. When heterogeneity is fully accounted for, the total incremental value of all observed ads is lower than the number of conversions, implying that many conversions would have occurred absent online advertising. Overall, the results highlight the importance of early-stage effects and show that rule-based attribution methods heavily overweight lower-funnel interactions.

2.4 Flaws and limitations

The study's reliance on a single automotive campaign limits external validity. Automotive purchases involve high deliberation, relatively long decision cycles, and particular information search patterns, so the channel roles observed here may not generalize to fast-moving consumer goods or services with shorter funnels. Search exposure is measured only through clicks, because search impressions are not observed; this restriction may understate or misrepresent the influence of non-clicked search ads and biases the relative contribution of search versus display.

On the modeling side, advertising exposures are treated as exogenous regressors: there is no explicit representation of the firm's decision problem or potential endogeneity arising from targeted ad delivery and bidding strategies. Offline media exposures and broader brand equity effects are absorbed into random effects rather than modeled jointly, leaving open questions about cross-media interactions. Finally, the assumption that all consumers begin disengaged is reasonable for a new product launch but may not hold for established brands with pre-existing awareness and loyalty, which could alter the inferred role of upper-funnel impressions.

2.5 Contributions and strengths

A central strength of the paper is its integration of a funnel-based behavioral structure with a flexible HMM estimated on rich, high-frequency user-level data. The latent-state formulation

provides a realistic representation of consumer deliberation and allows the model to capture both unobserved heterogeneity and state dependence in a coherent way. Empirical comparisons show that the four-state model significantly outperforms simpler alternatives, supporting the need for multiple engagement.

Substantively, the decomposition of advertising effects into state transitions and state-conditional conversion probabilities clarifies the mechanisms through which different media formats operate. The results indicate that display impressions primarily induce movement out of the disengaged state, while clicks and search activity exert their influence later in the process. The attribution analysis demonstrates that a structural, model-based approach can reweight the contribution of channels in ways that differ sharply from conventional metrics: search clicks lose share, while early display impressions gain. For this IS, the paper's main strength is to provide a concrete, empirically grounded example of stage-specific ad effects in a real multichannel campaign.

2.6 Relevance

Both this article and this IS share the premise that advertising works through a multi-stage funnel, with different activities influencing different stages. However, the goals diverge. Abhishek et al. aim to estimate how exposures affect consumer progression and to derive an attribution rule based on these dynamics; this IS focuses instead on optimal budget allocation across stages for a firm with stage-specific response functions and an explicit profit-maximization objective.

Despite this difference, the findings in Abhishek et al. provide empirical support for key assumptions in this IS. The stage-specific effects of display and search advertising validate modeling upper- and lower-funnel investments separately rather than combining them into a single advertising input. The definition of attribution value as an incremental effect on eventual conversion aligns closely with the marginal effectiveness term appearing in a Dorfman–Steiner-type allocation rule. The paper also shows that ignoring funnel dynamics leads to systematic underinvestment in early stages, strengthening the motivation for this IS's focus on stage-level marginal returns that depend on both firm characteristics and funnel position.

3. Conclusion

Media Exposure Through the Funnel provides a clear and empirically grounded framework for understanding how advertising shapes consumer movement through a multi-stage purchase process. By combining a latent-state model with detailed user-level data, the paper shows that different media channels operate at distinct points in the funnel and that their incremental contributions are often mismeasured by rule-based attribution methods such as last-touch. The results highlight the importance of modeling sequential progression when evaluating advertising effectiveness and offer a structured approach for reassessing the relative value of display and search channels.

For this IS, the key lessons are that (i) stage-specific advertising effects are real and economically meaningful, (ii) attribution should be based on incremental impact on eventual conversion rather than simple touch rules, and (iii) failing to account for funnel dynamics biases evaluation and, ultimately, budget allocation. These insights reinforce the conceptual foundations of the IS and support the development of a multi-stage, firm-contingent allocation model that builds explicitly on stage-specific marginal returns.

Article 3: Smith (2020) – Determinants of marketing budget allocation across Japan, Germany, and the United States

1. Introduction

This article provides an empirical bridge between the independent study's theoretical model and real-world evidence on how firm characteristics shape marketing outcomes. While the independent study asks how attributes like size, age, industry, and financial position should influence optimal budget allocation across funnel stages, Smith (2020) examines which organizational, regional, and performance variables are associated with high marketing profitability (net marketing contribution) for retail firms in Japan, Germany, and the United

States. The paper shows that factors such as regional business cycle, product price position, sales growth, liquidity, and the evolution of the marketing budget-to-sales ratio systematically distinguish firms with “maximum” marketing contribution from those with weaker performance.

This directly supports the independent study in three ways. First, it strengthens the theoretical foundation by confirming that firm characteristics and financial conditions matter for marketing effectiveness, which is a prerequisite for modeling firm-specific funnel responses. Second, the documented cross-country differences in dominant determinants, distinct patterns for Japan, Germany, and the US, reinforce the assumption of heterogeneity and justify treating optimal allocation as contingent on context rather than universal. Third, by grounding its argument in theories of the firm, bounded rationality, and regional business cycles, the paper helps connect the independent study’s economic and game-theoretic perspective to realistic managerial decision-making about marketing budgets.

2. Detailed Critical Description and Analysis

2.1 Article Overview and Objectives

Smith (2020) asks which firm- and context-level determinants are associated with high marketing profitability, measured by net marketing contribution (NMC), and whether these determinants are common across, or specific to, Japan, Germany, and the United States. The core questions are: (a) does a common set of “high-ranking” determinants of maximum NMC exist across all three countries, and (b) does each country exhibit its own unique set of dominant determinants.

The study uses cross-sectional data on 770 independent retail trade firms (NAICS 44–45 equivalents) active for at least five years, drawn from national business databases with associated secondary financial and marketing data collected in 2017. The sample includes 220 firms from Japan, 258 from Germany, and 292 from the United States. Firms are classified into three performance categories based on NMC as a percentage of sales: minimal (<10%), moderate (10–30%), and maximum (>30%).

The dependent variable is categorical NMC performance tier, and the author includes 21 independent variables grouped into three blocks: managerial (e.g., years in business, marketing budget-to-sales ratio, 1/3/5-year changes in that ratio), regional/market context (e.g., product price position, regional business cycle, domestic market share, number of locations, product class, number of competitors, breadth of product offering), and performance/financial measures (e.g., ROI, ROA, sales-to-inventories, inventory turnover, current ratio, sales growth, average product margin). The main hypotheses (P1–P3) are that there is a common cross-country determinant set for maximum NMC, that each country has a unique determinant subset, and that a probabilistic neural network will classify firms' NMC categories more accurately than linear discriminant analysis.

2.2 Methods and key equations

The paper uses a probabilistic neural network (PNN) to classify firms into three NMC performance tiers based on the 21 organizational, regional, and performance variables, and compares its accuracy to linear multiple discriminant analysis. A PNN is a non-parametric classifier rooted in Bayes decision theory: it estimates class-specific probability density functions (pdfs) from the data, then assigns each observation to the class with the highest estimated posterior probability.

Formally, for each class k (here: minimal, moderate, or maximum NMC), the PNN approximates the pdf $f_k(x)$ of the input vector x using Parzen window (kernel) density estimation:

$$f_k(x) = \frac{1}{(2\pi)^{d/2} s^d} \cdot \frac{1}{N_k} \sum_{i=1}^{N_k} \exp \left(-\frac{(x - x_{ki})^\top (x - x_{ki})}{2s^2} \right),$$

where x_{ki} is the i -th training example in class k , N_k is the number of examples in class k , d is the number of inputs, and s is a smoothing parameter. The classifier then uses Bayes' rule with class prior $p_k = N_k/N$ to choose the class with the highest posterior probability, operationalized as:

$$\sum_{i=1}^{N_k} \exp \left(-\frac{(x - x_{ki})^\top (x - x_{ki})}{2s^2} \right) \geq \sum_{i=1}^{N_j} \exp \left(-\frac{(x - x_{ji})^\top (x - x_{ji})}{2s^2} \right) \text{ for all } j \neq k.$$

Architecturally, the network has four layers: an input layer (21 neurons for the independent variables), a pattern layer (one neuron per training example computing the kernel), a summation layer (summing kernels within each class), and an output layer that implements the argmax decision rule. Performance categories are encoded as continuous targets (0.25 for minimal, 0.50 for moderate, 0.75 for maximum NMC), and the network is trained on 70% of observations with a learning rate of 0.7, momentum of 0.9, and early stopping based on validation error; the remaining 30% form a holdout validation set.

Feature extraction is performed by examining the normalized importance weights of each input in the trained PNN, which indicate how strongly small changes in a variable affect the classification decision. Variables with the largest normalized weights (summing to 1 across all inputs) are interpreted as “dominant determinants” of being in the maximum NMC class, both in the pooled three-country model and in country-specific models.

2.3 Results and interpretation

The PNN achieves very high classification accuracy, correctly assigning roughly 90% of firms in the combined three-country sample and around 90–94% within each country in the holdout validation, indicating that the 21 variables jointly contain strong predictive information about NMC performance tiers. In contrast, linear discriminant analysis correctly classifies only about 60% of firms overall and 59–69% by country, which supports the claim that the relationship between firm characteristics and marketing profitability is highly non-linear and better captured by the PNN.

In the pooled three-country model, feature extraction identifies five dominant determinants for belonging to the “maximum” NMC group: regional business cycle (RBC), product price position (RPP), firm sales growth rate (PGR), current ratio (PCR), and the five-year change in the marketing budget-to-sales ratio (OB5). This implies that high-NMC firms tend to operate in

more favorable regional economic conditions, occupy specific price positions (often premium or clearly defined), experience faster sales growth, maintain stronger short-term liquidity, and have actively adjusted their marketing budget intensity over the previous five years. These results support P1: there is a common cross-country set of high-ranking determinants of maximum NMC spanning managerial, regional, and financial dimensions.

Country-specific PNNs reveal additional, distinct dominant determinants, confirming P2 that each country has its own unique pattern. In Japan, short-term changes in the marketing budget-to-sales ratio (OB1) and the sales-to-inventories ratio (PSI) are particularly important, suggesting that agile budget adjustments and inventory productivity are critical for top marketing profitability. In Germany, firm asset size (OAS), domestic market share (RMS), and return on assets (PRA) emerge as dominant, indicating that larger, more established firms with strong market positions and asset efficiency are most likely to achieve maximum NMC. In the United States, the marketing budget-to-sales ratio (OBS), its three-year change (OB3), and firm sales growth rate (PGR) are key, reinforcing the view that both the level and trajectory of marketing intensity, together with growth, are central drivers of high marketing profitability in that context.

2.4 Flaws and limitations

A first limitation is that the study is cross-sectional and observational, so it cannot establish causal effects of the identified determinants on net marketing contribution; high NMC firms may differ on unobserved factors that jointly drive both the determinants and performance. Relatedly, all firms are retail trade companies, which limits external validity to other sectors (e.g., B2B, services, digital-first platforms) where budget allocation logic and constraints may differ substantially.

Second, while the PNN delivers high predictive accuracy, its “black box” nature makes structural interpretation difficult: the model ranks variables by importance but does not recover explicit elasticities or marginal effects that can be directly translated into optimal allocation rules, as in formal economic models. The discrete performance tiers (minimal, moderate, maximum NMC) also compress information about the underlying continuous profitability distribution, which may obscure more nuanced relationships between firm characteristics and marketing returns.

Third, the paper focuses on firm- and regional-level determinants but does not incorporate detailed information about how budgets are allocated across specific channels or funnel stages, nor about competitive interactions, which limits its ability to speak directly to multi-stage allocation problems like those in the independent study. Finally, the data are drawn from 2017 and from firms that have survived at least five years, which may introduce survivorship bias and reduce relevance for younger or more financially constrained firms.

2.5 Contributions and strengths

The paper's main contribution is to document, using a flexible non-linear classifier, that a small set of firm and contextual variables strongly predicts which firms achieve very high marketing profitability, and that this set has both common and country-specific components. By combining managerial variables (e.g., budget-to-sales levels and changes), regional conditions (business cycle, price position), and financial performance measures (growth, liquidity), the study offers a richer empirical picture of how marketing performance is embedded in the broader firm and market environment.

Methodologically, the use of a probabilistic neural network is a strength because it captures complex interactions and non-linearities that linear discriminant analysis cannot, as evidenced by the large improvement in classification accuracy (around 90% versus about 60%). The feature extraction step adds interpretive value by ranking determinants and distinguishing “dominant” drivers, so the analysis remains managerially meaningful despite the black-box nature of the model. Finally, the intra- and inter-country analyses are well designed: estimating both pooled and country-specific networks allows the paper to cleanly separate global regularities from local heterogeneity, which is conceptually aligned with cross-firm heterogeneity in the independent study.

2.6 Relevance

This article supports the independent study's core premise that firm characteristics systematically shape marketing effectiveness and, by implication, optimal allocation rules. By showing that variables such as sales growth, liquidity, product price position, and the trajectory of the marketing budget-to-sales ratio sharply distinguish high- from low-NMC firms, the paper

provides empirical backing for modeling stage-specific response parameters and budget constraints as functions of firm attributes rather than as constants. In other words, it helps establish that “who the firm is” and “where it sits” in the market materially affect the marginal returns to marketing, which is exactly the starting point needed before asking how those returns should translate into funnel-stage budgets.

The cross-country analysis reinforces the independent study’s emphasis on heterogeneity and contingency. The fact that Japan, Germany, and the United States exhibit different dominant determinants of maximum NMC. For example, short-run budget changes and inventory productivity in Japan, asset size and domestic share in Germany, and budget ratios and growth in the US, implies that even within a single broad sector, the mapping from spending to profitability is context dependent. This supports an allocation model in which optimal funnel-stage spending depends on firm size, age, industry, and macro environment, rather than a one-size-fits-all rule. Conceptually, the paper’s grounding in theories of the firm, bounded rationality, organizational learning, and regional business cycles complements the independent study’s economic and agency perspective, by highlighting how firm characteristics modify the constraints and incentives under which managers choose marketing budgets.

3. Conclusion

This article shows that a small, interpretable set of organizational, regional, and financial variables can accurately distinguish firms with very high net marketing contribution from those with weaker performance, and that these determinants include both common factors across countries and country-specific elements. From the perspective of the independent study, the key lessons are, first, that firm characteristics and financial conditions clearly matter for marketing effectiveness; second, that the mapping from marketing spend to profitability is heterogeneous across contexts, as illustrated by the different dominant determinants in Japan, Germany, and the United States; and third, that realistic models of budget allocation should embed these contextual differences in the response functions and constraints rather than assume a universal rule. Together, these insights provide empirical support and conceptual depth for treating optimal

multi-stage advertising allocation as contingent on firm attributes and market environment, in line with the theoretical framework developed in this IS.

Article 4: Wang et al. (2020) – Optimal advertising budget allocation across heterogeneous markets with multiple constraints

1. Introduction

“Optimal Advertising Budget Allocation across Markets with Different Goals and Various Constraints” by Wang et al. develops exactly the kind of optimization framework that this IS needs as a benchmark for multi-unit advertising allocation. While my IS focuses on allocating budgets across funnel stages and allowing firm characteristics to shape stage-specific effectiveness, Wang et al. study how a firm should optimally allocate a given advertising budget across multiple heterogeneous markets, each with its own response parameters and strategic goals. They build two explicit decision models: one that maximizes total sales subject to a fixed budget, and another that minimizes total advertising cost subject to a sales target, using the classic Vidale–Wolfe advertising response function as the behavioral core.

This paper is important for my IS for two reasons. First, it shows in a clear, mathematical way how to move from a dynamic response model to a constrained optimization problem, derive feasibility conditions, and solve for optimal advertising levels numerically. This is directly analogous to the step I take when I go from stage-specific logit response functions to a multi-stage allocation problem. Second, Wang et al. explicitly allow markets to differ in their “type” (new, share-building versus mature, profit-harvesting) and in their parameters, which provides a conceptual template for how different business segments or firm characteristics can lead to different advertising goals and constraints. Even though their heterogeneity is across markets rather than across funnel stages, the structure of their model and their main insights about feasibility, diminishing returns, and the equalization of marginal gains provide a useful theoretical complement to the funnel-based allocation framework developed in this IS.

2. Detailed Critical Description and Analysis

2.1 Article Overview and Objectives

Wang et al. ask how a firm should optimally allocate advertising budgets across multiple heterogeneous markets when each market has its own sales dynamics, strategic objective, and feasibility constraints. Specifically, they study two decision problems: (i) given a fixed total advertising budget, how to split it across markets to maximize total sales, and (ii) given a required total sales volume, how to achieve it at minimum advertising cost while respecting market-specific goals.

The paper is purely theoretical; there is no econometric estimation or real dataset. Instead, the “data” are model parameters and a numerical example with six hypothetical markets that differ in Vidale–Wolfe parameters (decay rate, saturation level, responsiveness), profit per unit, and target market share or profit. The key variables are the sales rate $s_i(t)$ in market i , the advertising expenditure rate $a_i(t)$, the decay constant θ_i , saturation level M_i , response constant r_i , the constant advertising rate c_i when spending is even over time, total sales S_i , per-unit profit p_i , target market share z_i for “new” (Type A) markets, and target profit ℓ_i for “mature” (Type B) markets.

The central hypotheses are that (1) using the Vidale–Wolfe response model, one can characterize for each market the minimal advertising needed to reach a share or profit target and the profit-maximizing advertising level, and (2) combining these market-level properties with a budget or sales constraint yields tractable nonlinear programming models that deliver optimal cross-market allocations.

2.2 Methods and key equations

The paper builds its decision models on the Vidale–Wolfe advertising response function. In a single market, the rate of sales $s(t)$ responds to advertising according to the differential equation

$$\frac{ds}{dt} = -\theta s(t) + r a(t)(M - s(t)),$$

where $\theta > 0$ is the sales decay constant, $M > 0$ is the market saturation level, $r > 0$ is the advertising response constant, and $a(t)$ is the advertising expenditure rate at time t .

For tractability, they assume that during a campaign of length τ advertising is spent at a constant rate c , so $a(t) = c$ over the campaign and then drops to zero afterwards. Solving the differential equation yields closed-form expressions for the sales rate at the end of the campaign, $s(T)$, and for total sales S over a horizon $[t, T]$ as functions of c, θ, M, r ; these expressions are then used to impose market-specific share and profit constraints.

For a “new” Type A market with target market share z , they define

$$F(c) = \frac{s(T)}{M} - z,$$

the gap between realized and desired share. They prove that $F(c)$ is strictly increasing in c when $r > 0$, and that a feasible target requires z to be below a market-specific upper bound; under this condition, the equation $F(c) = 0$ has a unique solution c^* , interpreted as the minimal advertising rate needed to hit the share goal.

For a “mature” Type B market with per-unit profit p and profit target ℓ , they define total profit as

$$G(c) = p S(c) - c,$$

where $S(c)$ is total sales from the Vidale–Wolfe solution. They show that $S(c)$ is increasing and concave in c , and that under reasonable parameter values $G(c)$ increases and then decreases, so there is a unique $c^\#$ that maximizes profit. They derive conditions under which the profit target ℓ is unattainable, trivially satisfied without advertising, or feasible; in the feasible case, solving $G(c) = \ell$ yields an interval of admissible advertising levels, with the lower root being the minimum-cost way to achieve the target.

These single-market properties are then embedded in two cross-market optimization models.

Let c_i be the advertising rate in market i , $S_i(c_i)$ total sales in that market, and C the total budget.

Model I (maximize sales given budget) is

$$\max_{c_i} \sum_i S_i(c_i)$$

subject to

$$\begin{aligned} \sum_i c_i &\leq C, \\ \frac{S_i(T)}{M_i} &\geq z_i \forall i \in A, \\ p_j S_j(c_j) - c_j &\geq \ell_j \forall j \in B, \end{aligned}$$

and $c_i \geq 0$. Using the monotonicity results, the share and profit constraints are transformed into linear bounds $c_i \geq c_i^{\min}$ or intervals $c_j^{\min} \leq c_j \leq c_j^{\max}$, so Model I becomes a nonlinear objective with linear constraints that can be solved numerically (e.g. in LINGO).

Model II (minimize cost given sales) is

$$\min_{c_i} \sum_i c_i$$

subject to

$$\sum_i S_i(c_i) \geq \bar{S},$$

together with the same transformed market-specific constraints and $c_i \geq 0$. This yields a linear objective with nonlinear constraints in $S_i(c_i)$, again solved as a nonlinear program.

2.3 Results and interpretation

The main analytical results establish feasibility conditions and uniqueness properties that make the optimization problems well behaved. For Type A markets, Wang et al. show that the function $F(c) = s(T)/M - z$ is strictly increasing in advertising c , so there is at most one advertising level that exactly attains a given target share z . They also derive an upper bound on feasible z : if the desired market share exceeds a threshold implied by θ, M, r , no amount of

advertising can hit the target, whereas for any feasible z there exists a unique minimal c^* that delivers $s(T)/M = z$.

For Type B markets, they show that total sales $S(c)$ increase with c but at a decreasing rate, and that profit $G(c) = pS(c) - c$ rises and then falls, implying a unique profit-maximizing advertising level $c^\#$. This leads to three regimes for a profit target ℓ : if ℓ is below the profit attainable with zero advertising, additional advertising is wasteful; if ℓ exceeds the maximum profit $G_{\max}$, the target is infeasible; and if ℓ lies between these bounds, there are exactly two solutions to $G(c) = \ell$, with the smaller root being the minimum-cost way to reach the target.

In the numerical example, a firm “XYZ” allocates a 7-million-dollar budget across six markets, two of which are Type A (share targets), two Type B (profit targets), and two unconstrained. Applying the theorems, the authors compute minimal advertising levels to satisfy each market’s share or profit constraint (for instance, roughly 0.76 and 0.65 million for the two new markets, and an interval $[0.63, 254.6]$ million for the profit-constrained mature market). Solving Model I with these transformed constraints yields an optimal allocation that fully uses the budget and produces total sales of about 25.14 million units across all markets, with some markets (e.g. a mature market whose profit target is already met without advertising) receiving zero spend and others receiving more than their minimum levels. The example illustrates how the theory can identify infeasible targets, compute rational thresholds for advertising in each market, and then choose a cross-market allocation that maximizes aggregate performance subject to those constraints.

2.4 Flaws and limitations

A first limitation is that the analysis is entirely theoretical and based on the Vidale–Wolfe response function, without empirical estimation or validation of the key parameters θ, M, r from data. In practice, firms would need reliable estimates of these parameters for each market; if they are misspecified, the “optimal” allocations could be far from optimal. The Vidale–Wolfe form itself is restrictive, assuming a particular interaction between decay, saturation, and advertising that may not fit many modern digital contexts.

Second, the model treats each market as a separate unit with no cross-market spillovers or synergies, and assumes constant advertising intensity over the campaign horizon. This rules out dynamic scheduling questions and any interdependence across markets (for example, global brand effects), which are central in multi-channel, multi-market settings. Third, the framework focuses on product-market heterogeneity but does not incorporate firm-level characteristics or funnel stages; parameters are exogenous and fixed rather than functions of firm traits or customer-journey position, which limits its direct applicability to the firm-heterogeneity and multi-stage allocation questions in this IS

2.5 Contributions and strengths

The paper's main contribution is to provide a clear, optimization-driven framework for allocating advertising budgets across multiple heterogeneous markets with different strategic goals and constraints. By embedding the Vidale-Wolfe response model into two decision problems, maximizing sales given a budget and minimizing budget given a sales target, the authors move beyond response estimation to explicit, solvable allocation rules. The derivation of feasibility conditions for market-share and profit targets, and the proofs of uniqueness for critical advertising levels c^* and $c^\#$, give the model theoretical discipline and make it implementable once parameters are known.

Methodologically, the paper is strong in how it decomposes the problem. It first characterizes single-market behavior (monotone and concave sales response, unimodal profit) and then uses these properties to transform a complex multi-market, multi-constraint problem into standard nonlinear programming models that can be solved numerically. The numerical example is transparent and shows how to check target feasibility, compute market-specific bounds, and use off-the-shelf software (LINGO) to obtain an optimal allocation. For this IS, the paper is valuable as a template for linking a dynamic response function to a constrained allocation problem and for illustrating classic economic ideas, diminishing returns and equalization of marginal gains, within a multi-unit advertising context.

2.6 Relevance

This paper is directly relevant to my IS because it shows, in a clean mathematical setting, how to move from an advertising response model to an explicit budget-allocation problem with clear optimality conditions. In Wang et al., the Vidale–Wolfe function plays the same role that stage-specific logit functions play in this IS: it links advertising intensity to sales, and its properties (monotonicity, concavity, thresholds) are then used to derive feasible regions and optimal allocations under constraints. Their results about minimal advertising needed to hit market-share or profit targets, and about profit-maximizing advertising levels, mirror the idea in this IS that optimal funnel-stage spending equalizes marginal profit per dollar across stages, subject to firm- and stage-specific response parameters.

Conceptually, the distinction between Type A (share-building) and Type B (profit-harvesting) markets is also useful. It provides a formal example of how differences in “development stage” and strategic objectives change the structure of the allocation problem and the constraints managers face. In my IS, firm characteristics such as age, size, or market position can be interpreted as shifting firms between share-building and profit-harvesting regimes at different funnel stages; Wang et al.’s treatment of heterogeneous markets with different goals offers a ready-made template for how such regime differences can be incorporated into an optimization model.

3. Conclusion

Wang et al. provide a rigorous example of how to embed a dynamic advertising response model in an explicit, multi-market budget allocation framework, derive feasibility conditions for different goals, and solve for optimal advertising levels under realistic constraints. The paper shows that minimal advertising needed to reach share or profit targets, and profit-maximizing advertising levels, can be characterized cleanly once response parameters are known, and that these market-level properties can be combined into tractable nonlinear programming models. For this IS, the key lessons are that (i) the path from response functions to allocation rules requires careful analysis of monotonicity, concavity, and feasibility; and (ii) heterogeneity in goals and

market conditions fundamentally shapes optimal allocation, reinforcing the need to model firm and funnel-stage differences explicitly rather than relying on a single, one-size-fits-all allocation rule.

Article 5: Chen et al. (2023) – Multichannel advertising budget allocation with spillover and carryover effects

1. Introduction

This article by Chen et al. addresses a central problem that also lies at the heart of this IS: how to allocate a fixed advertising budget across multiple channels when ads interact over time and across channels. Whereas my IS focuses on allocation across funnel stages and firm types, this paper studies allocation across channels (such as display and search) in the presence of spillover effects (one channel makes others more effective) and carryover effects (ads build a stock of goodwill that persists). The authors show that if a firm ignores these latent effects and relies on last-touch-style metrics to update budgets, the allocation converges to a stable but significantly suboptimal pattern. This directly challenges the common industry belief that repeated “optimize-on-attribution” is sufficient for good allocation.

For my IS, the paper is important in three ways. First, it provides a clear, structural example of how ignoring cross-channel and intertemporal interactions leads to systematic underinvestment in upper-funnel channels, even when managers use seemingly rational rules based on observed conversions. Second, it offers a tractable dynamic model—built on engagement and adstock state variables—that bridges attribution and optimization, showing how a response model with spillovers can be turned into an allocation rule. Third, the result that a simple, state-aware static budget split is nearly optimal for long campaigns suggests that sophisticated dynamics can sometimes be summarized by stable allocation rules, a theme that aligns with the Dorfman–Steiner-type conditions and multi-stage allocation principles developed in this IS.

2. Detailed Critical Description and Analysis

2.1 Article Overview and Objectives

Chen et al. ask how a marketing agency should allocate a fixed advertising budget dynamically across multiple channels when there are spillover and carryover effects, and what happens if the agency instead follows the common last-touch, “channel-in-silos” approach. Their core questions are: (i) how large is the performance loss from ignoring latent cross-channel interactions, and (ii) what structure do optimal policies have when these effects are modeled explicitly.

The paper combines a theoretical model with an empirical illustration. The main analysis is based on a fluid (aggregate) model of a large population of customers exposed to N channels over a time horizon $[0, T]$. Key variables include the budget rate by channel $b_n(t)$, the total budget rate B , the impression production function $f_n(b_n(t))$ that maps spend to impressions, and latent customer state variables: the fraction of engaged customers $x^e(t)$ and the average adstock (accumulated goodwill) $x^s(t)$. Channel-specific parameters ϵ_n and σ_n govern how impressions increase engagement and adstock, and a conversion-rate function $p_n(x(t))$ links the current state to the probability that an impression on channel n generates a conversion. In addition, the authors use daily aggregate data from a six-month online fashion campaign (spend, impressions, clicks, and conversions by channel) to estimate the model parameters and compare policies via simulation.

The main hypotheses are that (1) state-oblivious budget rules that ignore latent effects and repeatedly reallocate based on observed per-channel conversion rates will converge to a stationary but suboptimal allocation, losing a non-trivial share of potential conversions, and (2) a state-aware allocation that internalizes spillover and carryover effects has a simple “turnpike” structure and can be well approximated by a static split that is nearly optimal for long campaigns.

2.2 Methods and key equations

The authors build a dynamic, multichannel response model with two latent state variables: the fraction of engaged customers $x^e(t)$ and the average adstock (cumulative goodwill) $x^s(t)$. The

firm allocates a fixed total budget rate B across N channels, choosing $b_n(t)$ with $\sum_{n=1}^N b_n(t) = B$ at each time t . Spending $b_n(t)$ on channel n generates impressions at rate $f_n(b_n(t))$, where f_n is increasing and concave.

In the fluid (large-market) limit, the evolution of the latent states is described by a system of ordinary differential equations. Engagement increases when disengaged customers see impressions and decreases when engaged customers lose interest:

$$\dot{x}^e(t) = -\eta x^e(t) + (1 - x^e(t)) \sum_{n=1}^N \epsilon_n f_n(b_n(t)),$$

where η is the disengagement rate and ϵ_n measures how effectively channel n creates engagement. Adstock decays over time but rises when engaged customers see impressions:

$$\dot{x}^s(t) = -\sigma x^s(t) + x^e(t) \sum_{n=1}^N \sigma_n f_n(b_n(t)),$$

where σ is the decay rate and σ_n captures how much goodwill an impression on channel n adds.

Given the state $x(t) = (x^e(t), x^s(t))$, the aggregate conversion rate on channel n at time t is $p_n(x(t))f_n(b_n(t))$, where $p_n(\cdot)$ is an increasing function reflecting that higher engagement and adstock raise conversion propensity. The fluid objective is to maximize expected total conversions over the campaign:

$$\max_{b(\cdot)} J^T = \int_0^T \sum_{n=1}^N p_n(x(t)) f_n(b_n(t)) dt$$

subject to the two ODEs above and the budget constraint $\sum_n b_n(t) = B$.

This is a continuous-time optimal control problem. Using Pontryagin's maximum principle, the authors derive first-order conditions that balance, for each channel, the direct marginal impact on current conversions and the indirect marginal impact on future engagement and adstock. They show that optimal policies have a “turnpike” structure: after an initial “bump-up” phase to build

state and a final “exploitation” phase, the optimal allocation is approximately constant for most of the horizon.

To model industry practice, they also define a state-oblivious policy: the horizon is split into phases, and at the start of each phase the agency estimates per-channel conversion rates from the previous phase and solves a static problem

$$\max_{b \in \mathcal{B}} \sum_{n=1}^N \hat{p}_n f_n(b_n),$$

ignoring the latent states. This policy mimics last-touch budgeting and serves as a benchmark for quantifying the performance loss relative to the state-aware optimal and near-optimal static policies.

2.3 Results and interpretation

The first main result is that state-oblivious, last-touch–style budgeting converges to a stable but suboptimal allocation when spillover and carryover effects exist. Because observed per-channel conversion rates are themselves shaped by past budget decisions through the latent state, repeatedly reallocating toward channels with high measured conversions creates a feedback loop: channels that build engagement and adstock but do not get direct last-click credit are gradually underfunded, which further reduces their measured conversions and pushes the allocation even further away from the true optimum. In numerical experiments with two channels, this dynamic leads to long-run budgets that deliver only about two-thirds of the conversions achievable under the state-aware optimum, showing that the “price of misattribution” can be large.

The second main result characterizes optimal policies in the full state-aware model. Using the fluid formulation, the authors show that optimal dynamic allocations exhibit a three-phase “turnpike” structure: an initial bump-up phase that invests more in channels that efficiently build engagement and adstock, a long stationary phase where both latent states and budget shares are approximately constant, and a final exploitation phase that shifts budget toward channels with high immediate conversion productivity. For sufficiently long campaigns, almost all value is

generated in the stationary phase, so a static allocation that coincides with the stationary shares is provably near-optimal, with a performance gap that shrinks at rate $1/T$.

Finally, using data from a six-month multichannel campaign of a large online fashion retailer, the authors estimate the model parameters and simulate counterfactual policies. In this empirical setting, the state-aware static policy achieves conversions that are very close to the fully optimal dynamic benchmark, while the state-oblivious last-touch policy performs noticeably worse, especially in longer campaigns. The empirical results confirm that properly accounting for latent cross-channel effects can materially improve total conversions, and that a simple, well-designed static split can capture most of these gains.

2.4 Flaws and limitations

A first limitation is that the model is highly stylized and operates at an aggregate, fluid level, assuming a very large customer base and smooth state dynamics. This abstraction ignores individual-level heterogeneity in responsiveness and path-to-purchase, and it assumes that engagement and adstock can be summarized by just two continuous state variables for the entire market, which may be too coarse for some applications.

Second, the functional forms and parameters for impression production $f_n(\cdot)$, state dynamics, and conversion rates $p_n(\cdot)$ are not derived from microfoundations but imposed for tractability. Mis-specification of these functions could materially affect the recommended allocations, and estimation is done on a single campaign in one industry, limiting external validity. Third, the framework focuses on cross-channel and intertemporal effects but does not incorporate firm-level characteristics, competitive reactions, or explicit funnel stages, which makes the link to firm heterogeneity and stage-specific allocation, central themes in this IS, conceptual rather than direct.

2.5 Contributions and strengths

The paper's main strength is its clear, structural demonstration of how ignoring spillover and carryover effects can systematically distort multichannel budget allocation. By explicitly modeling engagement and adstock dynamics and then contrasting a state-aware policy with a

state-oblivious, last-touch–style rule, the authors turn a vague concern about “misattribution” into a precise, quantifiable performance loss: a sizable fraction of potential conversions is left on the table when latent effects are ignored.

Methodologically, the fluid formulation is a strong contribution. It reduces an intractable stochastic dynamic programming problem with many customers to a deterministic control problem in two state variables, enabling analytical characterization of optimal policies and the proof of a turnpike structure. This, in turn, leads to a practically appealing result: a simple static allocation that accounts for latent effects is nearly optimal for long campaigns, offering a tractable bridge between theory and practice. The empirical illustration, though limited in scope, shows that the model can be estimated from aggregate campaign data and that the proposed policy delivers meaningful gains over naive last-touch budgeting in a real multichannel setting.

2.6 Relevance

This paper reinforces the central argument of this IS that advertising decisions cannot be made by treating channels or stages in isolation. By showing that last-touch–style rules systematically underinvest in channels that mainly build engagement and adstock, Chen et al. provide a concrete mechanism for why upper-funnel or assistive activities tend to be undervalued when firms rely on naive attribution. This directly parallels my concern that stage-wise allocation based only on observed conversions will bias spending toward lower-funnel actions and away from awareness or consideration investments.

Conceptually, the engagement adstock state variables in their model play a similar role to funnel stages in my framework: they capture how current advertising affects future responsiveness and thus the marginal value of later actions. The turnpike result, that a simple, state-aware static allocation can approximate the fully optimal dynamic policy, also aligns with the Dorfman–Steiner-type logic in this IS, where optimal rules equate marginal profit per dollar across inputs but can be summarized by stable ratios. Finally, by focusing on cross-channel spillovers and carryovers, the paper complements my emphasis on firm and stage heterogeneity: together, they motivate an allocation model in which both firm characteristics and interdependencies across stages/channels shape the true marginal returns to advertising

3. Conclusion

From this article, I learn that cross-channel spillover and intertemporal carryover are not minor corrections but core determinants of optimal multichannel budget allocation. When these latent effects are ignored and budgets are updated using last-touch–style conversion metrics, the allocation converges to a stable yet substantially suboptimal mix that systematically underfunds channels whose main role is to build engagement and adstock. In contrast, a state-aware policy that internalizes these dynamics has a simple three-phase structure and, for long campaigns, can be well approximated by a static budget split that equalizes marginal returns once latent effects are considered.

For this IS, the key takeaway is that any theory of funnel-stage allocation must recognize interactions across stages and over time, not just stage-specific elasticities. The engagement–adstock framework offers a tractable way to represent these interactions, and the turnpike result suggests that complex dynamics can still yield relatively simple, stable allocation rules, strengthening the case for the multi-stage, firm-contingent allocation conditions developed in my theoretical model.

Empirical Proposal

A direct empirical test of “How do firm characteristics shape the optimal allocation of advertising budgets across funnel stages?” is constrained by a core data problem: firms don’t disclose how their total advertising budget is split between upper-funnel (brand building/awareness) and lower-funnel (activation/conversion) activities in public filings. As a result, even when total advertising expense can be collected from firms’ Form 10-K disclosures, the within-budget allocation across funnel stages or even advertising channels are typically unobserved.

This proposal constructs an imputed (not observed) upper-funnel budget share by combining (i) firm-year total advertising expense from 10-Ks with (ii) an evidence-based benchmark for

effective long-run marketing investment. Binet and Field (2013) argue that, on average, effective marketing balances long-term brand building and short-term activation at approximately 60/40 (brand/activation), while emphasizing the ratio can vary with context. In addition, the motivating logic for allowing the split to vary with firm scale is that smaller advertisers often have limited capacity to run upper-funnel campaigns and tend to concentrate more on lower-funnel outcomes, whereas larger advertisers have more scope to invest in brand-building objectives [Deisenroth et al., 2024].

Data construction: total advertising budget

To construct a firm-year panel of total advertising budgets, the empirical sample will focus on public companies with readily accessible annual reports (Form 10-K for U.S. filers, and comparable annual-report filings for non-U.S. issuers/ADRs when applicable). The initial sampling frame includes large, advertising-intensive consumer and telecom/media firms (e.g., Procter & Gamble (PG), PepsiCo (PEP), The Coca-Cola Company (KO), Unilever PLC (UL), L’Oréal (ADR: LRLCY), Nestlé (ADR: NSRGY), Comcast (CMCSA), AT&T (T), Verizon (VZ), Amazon (AMZN)) and a set of smaller public firms to ensure variation in scale (Wilhelmina International (WHLM), Direct Digital Holdings (DRCT), Dutch Bros (BROS), Interactive Strength (TRNR), Twin Hospitality Group (TWNP), First Community Corporation (FCCO), Northeast Community Bancorp (NECB), Big 5 Sporting Goods (BGFV)).

For each firm i and fiscal year t , collect total advertising expense B_{it} from Form 10-K disclosures and related notes (when presented).

This produces a firm-year panel of total advertising budgets, but not funnel shares.

Firm size classification

Because the imputation rule uses scale, “firm size” is operationalized using the log of advertising spend:

$$x_{it} = \log (B_{it})$$

This log transform is used because advertising spend is highly skewed and log scaling stabilizes the measure of “size” in relative (percentage) terms.

To interpret “small vs large” firms descriptively (e.g., in summary tables), firms can also be assigned to size bins each year based on the distribution of B_{it} (e.g., Small = bottom quartile, Medium = middle two quartiles, Large = top quartile), while the main imputation uses the continuous measure.

Standardizing size (so “median firm” is the reference)

Define standardized size:

$$z_{it} = \frac{x_{it} - \text{median}(x_{\cdot t})}{\text{sd}(x_{\cdot t})}$$

This normalization makes $z_{it} = 0$ correspond to the median advertiser (by ad spend) in year t , enabling a transparent calibration of the baseline split.

Imputing the upper-funnel share

Let $s_{it} \in (0,1)$ denote the imputed upper-funnel budget share.

We map standardized size into a share using the inverse-logit (logistic) function, which ensures the imputed share is always between 0 and 1.

$$s_{it} = \text{invlogit}(\alpha + \beta z_{it}) = \frac{1}{1 + e^{-(\alpha + \beta z_{it})}}$$

Interpretation of α and β

- α is the intercept that pins down the upper-funnel share when $z_{it} = 0$ (the median-size firm).
- β controls how strongly the upper-funnel share increases with size; $\beta > 0$ implies larger firms allocate a larger fraction to upper funnel.

Calibration to the 60/40 benchmark

To anchor the imputation to Binet and Field's 60/40 benchmark at the median firm, set α so that $s_{it} = 0.60$ when $z_{it} = 0$.

Since $\text{invlogit}(\alpha) = 0.60$, this implies:

$$\alpha = \log \left(\frac{0.60}{0.40} \right) \approx 0.405$$

Then β is treated as a design parameter and will be evaluated via sensitivity analysis (e.g., low/medium/high slopes) to ensure results are not driven by an arbitrary choice.

Converting shares into dollar allocations

Given B_{it} and s_{it} , compute:

- Upper-funnel dollars: $U_{it} = s_{it} \cdot B_{it}$
- Lower-funnel dollars: $L_{it} = (1 - s_{it}) \cdot B_{it}$

Main empirical specification

The main empirical analysis relates the constructed allocation measure to observable firm characteristics (beyond the size-based imputation).

A baseline regression is:

$$s_{it} = \theta_0 + \theta_1 \text{Age}_{it} + \theta_2 \text{InvTurn}_{it} + \theta_3 \text{AdvInt}_{it} + \theta_4 \text{RDInt}_{it} + \theta_5 \text{ROA}_{it} + \delta_{\text{industry}} + \tau_t + \varepsilon_{it}$$

where δ_{industry} are industry fixed effects and τ_t are year fixed effects.

Variable definitions

- s_{it} : imputed upper-funnel share of total advertising budget (fraction of B_{it} allocated to upper-funnel activities, constructed using the size-based rule).|
- Age_{it} : firm age in years (years since founding or IPO, from 10-K or Compustat).|

- InvTurn_{it} : inventory turnover ratio (cost of goods sold divided by average inventory, from 10-K income statement and balance sheet).|
- AdvInt_{it} : advertising intensity (advertising expense B_{it} divided by sales revenue, from 10-K).|
- RDInt_{it} : R&D intensity (R&D expense divided by sales revenue, from 10-K).|
- ROA_{it} : return on assets (net income divided by total assets, from 10-K income statement and balance sheet).|
- δ_{industry} : industry fixed effects
- τ_t : year fixed effects.

Important identification note: because s_{it} is mechanically constructed using B_{it} through z_{it} , the main regression focuses on other firm characteristics (age, inventory turnover, etc.) rather than re-including $\log(B_{it})$ on the right-hand side, to avoid a tautological relationship.|

Empirical Chapter

1. Data Presentation

1.1 Sample construction and context

The empirical analysis builds on a firm-level panel that I assembled manually from annual reports. The starting point is 18 large, publicly listed firms that operate in consumer-facing markets and rely heavily on brand advertising. The sample spans seven industries—fast-moving consumer goods (FMCG), technology, media/telecom, automotive, healthcare, retail, and leisure—so it captures very different business models and marketing environments. Within each industry I focus on a small set of flagship companies (for example, P&G, Unilever, Coca-Cola, and PepsiCo in FMCG), which makes it possible to follow the same firms over time and connect the numbers back to their actual strategies

For each firm, I collected yearly financial and operating data from its published annual reports, including the income statement, balance sheet, and notes. The core variables are advertising expenses, cost of goods sold (COGS), revenues, net income, inventories, total assets, total debt, cash on hand, research and development (R&D), average inventory, and average total assets. Because these variables are not available as a ready-made dataset, the construction of the panel required reading and extracting information from each annual report, and then harmonizing definitions across firms and over time. This “bottom-up” approach is more laborious than using a commercial database but has the advantage that the advertising and marketing-related quantities are measured as the firm itself reports them.

The resulting panel covers fiscal years 2010–2024, with some missing observations due primarily to gaps in inventory data. To construct lagged variables such as average inventory and average total assets, I initially extended the raw collection back one additional year to 2009. Year 2009 is used only as a “pre-sample” year to compute averages and lags; it is dropped before estimation so that all regressions are run on the 2010–2024 window. After applying these choices and excluding firm-years with missing key balance-sheet or advertising information, the main estimation sample consists of 270 firm-year observations across the 18 firms and 7 industries.

A practical complication concerns inventory data for Comcast. For this firm, inventory (and thus inventory turnover) is not reported in a way that can be made comparable to other sample firms. As a result, all Comcast observations are missing for the inventory-related measures. When I run specifications that include inventory turnover as an explanatory variable, these Comcast observations drop out and the effective sample shrinks from 269 to 254 usable firm-years. This mechanical difference in sample size is why I estimate both versions of each model, with and without inventory turnover.

Finally, I deliberately keep the variables close to their raw accounting form. Apart from taking logarithms (for firm size) and constructing lags, I do not winsorize extreme values or apply additional transformations. This choice keeps the empirical results as transparent as possible: rather than smoothing away outliers mechanically, I rely on the model structure (in particular, industry and firm effects) to absorb some of the heterogeneity. The downside is that the

regressions must cope with large differences across firms, but the upside is that any patterns that emerge can

Table 1.1 - Sample overview

Panel A: Sample Overview	
Number of Firms	18
Number of Observations	270
Years Covered	2010 to 2024
Number of Industries	7

1.2 Measuring upper-funnel share

The key outcome variable is the upper-funnel share (UFS), denoted s_{it} , which is meant to capture the fraction of a firm's advertising budget devoted to upper-funnel activities—awareness and consideration—in year t . Ideally, firms would disclose this split directly (for example, the share of spend on brand vs performance campaigns), but in practice such breakdowns are not systematically reported and no public database tracks funnel-level allocations

To get around this, I use the theoretical model from the previous chapter to impute the upper-funnel share from observed total advertising spend. The core idea from the theory is that optimal upper- vs lower-funnel spending should vary systematically with firm characteristics, in particular with firm size. I therefore treat the scale of advertising expenditure as a sufficient statistic for where a firm sits on the funnel spectrum.

The construction has three steps.

1. **Size from advertising spend.** For each firm-year I define

$$x_{it} = \log (B_{it}),$$

2. **Standardizing within each year.** To net out shifts in the overall level of ad spending over time, I express each firm's log-spend relative to its peers in the same year:

$$z_{it} = \frac{x_{it} - \text{median}(x_{.t})}{\text{sd}(x_{.t})}.$$

3. **Mapping size into a share.** Finally, I map standardized size into an imputed upper-funnel share using a logistic function:

$$s_{it} = \text{invlogit}(\alpha + \beta z_{it}) = \frac{1}{1 + e^{-(\alpha + \beta z_{it})}}.$$

The parameters α and β are calibrated so that the implied shares fall in a realistic range—for example, a median-sized firm allocates around half of its budget to the upper funnel, while very large advertisers tilt somewhat more in that direction. The logistic form guarantees $s_{it} \in (0,1)$ and produces the S-shaped response that the theory suggests

In the empirical part of the study, all references to UFS use this imputed share s_{it} ; there are no direct firm-reported upper-funnel shares hiding in the background

Table 1.2 – Dependent variable: Upper-funnel share

Panel B: Dependent Variable							
Variable	Mean	ST DEV	P25	Median	P75	Min	Max
Upper Funnel Share	0.55	0.09	0.45	0.60	0.62	0.33	0.68

1.3 Firm characteristics

The main explanatory variables are standard accounting ratios based on the same annual report data. All of them are lagged one year, so the financial position in year $t - 1$ is used to explain the allocation decision in year t . This timing matches the idea that budgets are set using information from the previous year rather than at the same time as the outcome

- **Return on assets (ROA_{it-1})**: is net income divided by average total assets, using beginning- and end-of-year assets for year $t - 1$
- **Leverage $_{it-1}$** : is total debt over total assets in year $t - 1$
- **Cash ratio $_{it-1}$** : is cash and cash equivalents divided by total assets in year $t - 1$
- **Firm size $_{it-1}$** : is the natural logarithm of total assets in year $t - 1$; this is the accounting counterpart to the size concept used in the UFS imputation
- **Inventory turnover $_{it-1}$** : is cost of goods sold divided by average inventory in year $t - 1$; for reasons discussed above, this is missing for all Comcast observations

All income-statement and balance-sheet items—advertising, COGS, revenues, net income, inventories, total assets, total debt, cash, and R&D—are taken directly from the annual reports. R&D was collected initially but dropped from the baseline regressions because it is missing too often to support a stable panel. Industry dummies assign each firm to one of the seven sectors listed above, and year dummies soak up macro shocks that affect all firms in a given year

Table 1.3 – Independent variables: summary statistics

Panel C: Independent Variables							
Variable	Mean	ST DEV	P25	Median	P75	Min	Max
ROA	0.09	0.06	0.05	0.10	0.13	-0.08	0.23
Cash Ratio	0.09	0.10	0.01	0.05	0.12	0.00	0.52
Leverage	0.27	0.19	0.12	0.25	0.41	0.00	0.64
Firm Size	7.86	1.54	6.63	7.74	9.24	4.34	10.35
Inventory Turnover	5.94	4.36	3.70	4.32	5.16	2.25	20.41

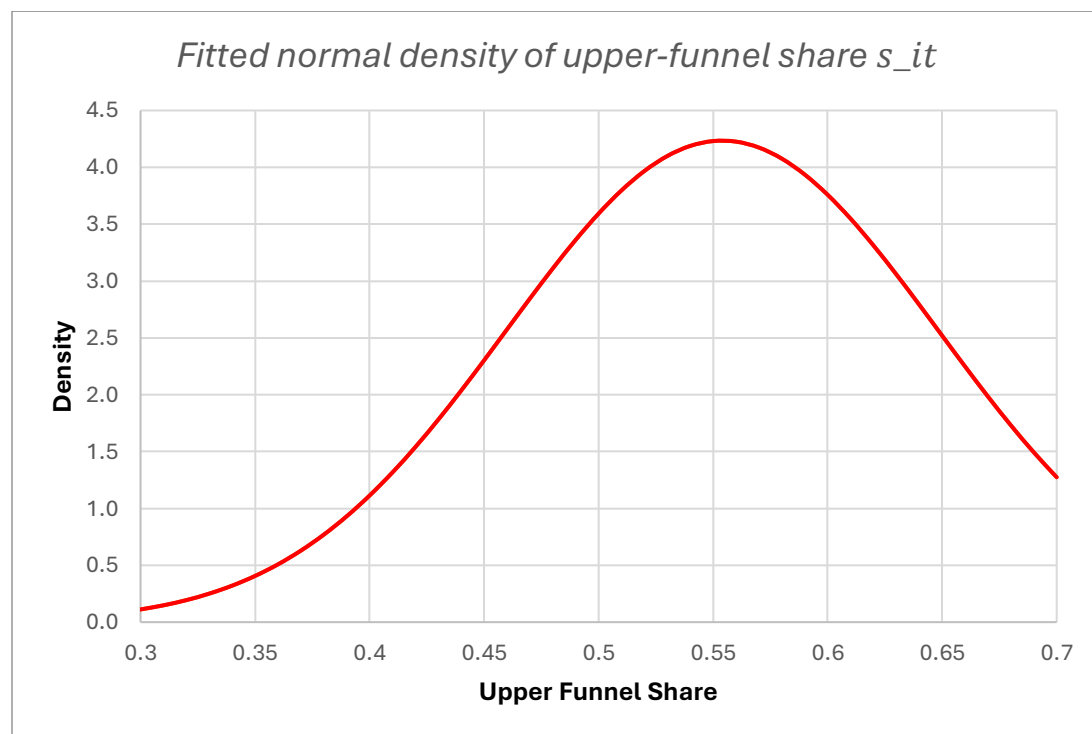
1.4 Descriptive patterns

Before moving to the regressions, it is helpful to see how the main variables behave in the raw data. Table 1.2 shows that the average upper-funnel share is about 0.55, with a standard

deviation of 0.09 and a range from roughly 0.33 to 0.68. So, in a typical year, the firms in the sample devote slightly more than half of their ad budgets to upper-funnel activity, but there is enough spread that some firms are clearly more lower-funnel-oriented and others are clearly more upper-funnel-oriented

Table 1.1 plots a histogram of UFS together with a fitted normal density. The distribution is close to bell-shaped: most observations lie between 0.45 and 0.63, and the densest region is in the 0.53–0.63 range. Very few firm-years allocate less than one-third or more than two-thirds of the budget to the upper funnel. This pattern matches the theoretical prediction that optimal allocations tend to be interior—firms almost never shut down either upper- or lower-funnel spending entirely

Table 1.3 – Histogram of UFS with fitted normal density



Panel C of Table 1.3 highlights the spread in financial characteristics. ROA averages roughly 9–10 percent but ranges from negative observations to years with very high profitability. Leverage has a mean around 0.27 and a fairly wide standard deviation, reflecting firms that are almost

unlevered alongside firms with substantial debt. Log total assets ranges from about 4.3 to just over 10, implying an order-of-magnitude difference in scale between the smallest and largest firms in the sample. Inventory turnover is also quite dispersed, with some firms turning inventory only a few times per year and others many times, consistent with differences between fast-moving consumer goods, retail, and more capital-intensive sectors

Even in these simple plots and tables there are patterns that the regressions will formalize. The largest firms by total assets tend to appear in the high-UFS portion of the distribution, while smaller firms are more common in the low-UFS bins. Likewise, industries with very fast inventory cycles or high COGS ratios look systematically different from slower-moving sectors in terms of typical UFS values. The next section sets up the empirical strategy used to separate these effects and test how firm characteristics and industry membership shape upper-funnel allocation

1.5 Industry composition and cross-sectional differences

The sample is deliberately spread across industries but remains focused on settings in which brand building and large-scale advertising campaigns are central to the business model. FMCG contributes four firms, media/telecom three, automotive and healthcare two each, retail five, and leisure one, alongside a single large technology platform. This mix gives enough cross-sectional variation to compare very different marketing environments while still keeping the number of firms manageable

Figure 1.2 plots the mean upper-funnel share by industry. The pattern is clear. FMCG, technology, media/telecom, automotive, and healthcare all allocate around 60–63 percent of their budgets to upper-funnel activity. Retail and leisure stand out on the low side, with average shares around 0.44 and 0.43 respectively. This fits basic intuition: FMCG brands and large platforms live or die by broad awareness and brand strength, whereas retailers and leisure firms tend to operate with thin margins and more tactical, conversion-oriented campaigns

Figure 1.2 – Average UFS by industry (bar chart)

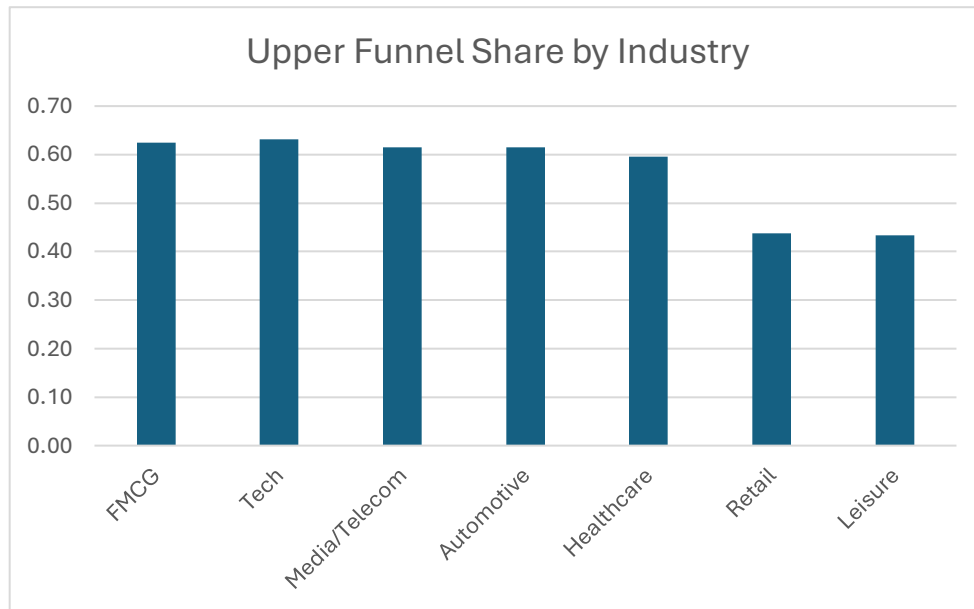


Table 1.4 reports industry-level averages of UFS and the main firm characteristics. Industries that lean heavily into upper-funnel spending—such as FMCG and tech—also tend to host the largest firms in terms of total assets. Retail and leisure are smaller on average and differ in leverage and profitability, which makes it hard to attribute their low UFS purely to “industry culture” without controlling for these other factors. This is exactly the motivation for the panel regressions in the next section: to disentangle the role of size, balance-sheet structure, and industry in shaping how firms allocate their advertising budgets across the marketing funnel

Table 1.4– Industry-level averages (UFS, ROA, cash, leverage, firm size)

Industry	Average				
	UFS	ROA	Cash	Leverage	Firm size
FMCG	0.62	0.10	0.09	0.36	11.35
Tech	0.63	0.04	0.28	0.20	11.71
Media/Telecom	0.61	0.20	0.05	1.38	11.84
Automotive	0.61	0.03	0.11	0.47	12.29
Healthcare	0.60	0.09	0.12	0.21	12.00

Retail	0.44	0.09	0.09	0.27	7.86
Leisure	0.43	0.04	0.05	0.84	7.77

2. Empirical strategy

The empirical goal is to connect the theory-driven upper-funnel share s_{it} to observable firm characteristics. The question is whether differences in profitability, liquidity, leverage, and size—together with industry and time effects—help explain why some firms allocate more of their advertising budgets to the upper funnel than others. This section describes how I translate that idea into regression models and why I estimate two complementary specifications (with and without firm fixed effects), each in versions that include or exclude inventory turnover.

At a high level, the strategy has three steps. First, I estimate a pooled model with industry dummies to measure systematic cross-sectional differences in upper-funnel allocation across industries, controlling for firm-level covariates and common time shocks (Model 1). Second, I move to a firm fixed-effects model to study within-firm dynamics—how changes in a given firm’s characteristics over time are associated with changes in its upper-funnel share (Model 2). Third, for both models I estimate a version that also includes inventory turnover, recognizing that the available sample shrinks because Comcast has no comparable inventory data.

2.1 Model 1: Industry differences in upper-funnel allocation

2.1.1 Motivation

The first model focuses on how upper-funnel allocation varies across industries. The theory chapter emphasized that firm characteristics, including industry and size, shape both the profitability of advertising and the optimal split across funnel stages. Industries differ systematically in business models, customer purchase cycles, and the role of brand equity. For example, FMCG firms rely heavily on broad awareness and shelf presence, while retailers and leisure businesses may lean more on promotions and conversions. Model 1 is designed to quantify these differences after conditioning on basic financial controls.

2.1.2 Specification

Model 1 is a pooled regression with industry and year fixed effects:

$$s_{it} = \beta_1 \text{ROA}_{i,t-1} + \beta_2 \text{Cash}_{i,t-1} + \beta_3 \text{Leverage}_{i,t-1} + \beta_4 \text{Size}_{i,t-1} + \lambda' D_i^{\text{ind}} + \tau' D_t^{\text{year}} + \varepsilon_{it},$$

estimated on the full sample of 269 observations (excluding the pre-sample year and any missing values). The vector D_i^{ind} contains dummies for FMCG, Tech, Media/Telecom, Automotive, Healthcare, Retail, and Leisure, with one category omitted as the reference industry. The coefficients on these dummies capture average differences in UFS between each industry and the omitted base, holding fixed ROA, cash, leverage, size, and year.

The extended version of Model 1 adds lagged inventory turnover:

$$s_{it} = \dots + \beta_5 \text{InvTurn}_{i,t-1} + \varepsilon_{it},$$

and is estimated on the reduced sample of 254 observations once Comcast drops out.

2.2 Model 2: Within-firm dynamics

2.2.1 Motivation

Model 1 uses cross-sectional variation and industry dummies to highlight level differences in UFS across industries and firm types. However, it does not fully exploit the panel dimension of the data. The second model asks a different question: when a given firm becomes more profitable, changes its leverage, or grows in size, does its upper-funnel share move in a predictable direction? To answer this, I use a firm fixed-effects specification that focuses on within-firm changes over time.

The logic is familiar from the theory chapter. The logit-based allocation model implies that the parameters governing stage-specific advertising effectiveness depend on firm characteristics such as size, industry, and “type.” Many of these characteristics are persistent. A firm’s business model, brand position, and internal capabilities are unlikely to change dramatically year to year. A pooled regression may therefore confound permanent, unobserved differences between firms with the effects of the observed regressors. By introducing firm fixed effects, Model 2 controls

for all time-invariant firm attributes—observed and unobserved—and zooms in on how changes in ROA, cash, leverage, and size are related to changes in UFS for the same firm.

2.2.2 Specification

Model 2 takes the form

$$s_{it} = \alpha_i + \beta_1 \text{ROA}_{i,t-1} + \beta_2 \text{Cash}_{i,t-1} + \beta_3 \text{Leverage}_{i,t-1} + \beta_4 \text{Size}_{i,t-1} + \tau' D_t^{\text{year}} + v_{it},$$

where α_i is a firm-specific intercept capturing all time-invariant differences between firms. The model is estimated using `xtreg, fe` with robust standard errors clustered by firm. The within R^2 of about 0.58 indicates that the time variation in these regressors explains a substantial portion of the within-firm variation in UFS.

The extended version adds inventory turnover:

$$s_{it} = \alpha_i + \beta_1 \text{ROA}_{i,t-1} + \beta_2 \text{Cash}_{i,t-1} + \beta_3 \text{Leverage}_{i,t-1} + \beta_4 \text{Size}_{i,t-1} + \beta_5 \text{InvTurn}_{i,t-1} + \tau' D_t^{\text{year}} + v_{it}.$$

Because Comcast drops out once inventory turnover is included, the sample for this version contains 254 observations across 17 firms.

The interpretation now focuses on dynamics. A positive β_4 tells us that when a given firm grows in size (measured by log assets), holding ROA, cash, leverage, and year fixed, its upper-funnel share tends to increase. Similarly, a positive β_3 means that when the same firm increases its leverage, its UFS also rises on average. Fixed effects strip out all level differences across firms and let us read these coefficients as responses to within-firm changes rather than cross-firm comparisons.

In all specifications I estimate the model by ordinary least squares or fixed-effects regression with robust standard errors clustered at the firm level, which accounts for arbitrary serial correlation and heteroskedasticity within each firm's time series. The Stata commands used are:

`reg` with industry and year dummies for Model 1.

`xtreg, fe` with year dummies and firm clustering for Model 2.

2.3 Choice of controls and estimation details

The set of controls is intentionally parsimonious. ROA, leverage, cash, and size capture four basic dimensions of the firm’s economic environment: profitability, financial risk, liquidity, and scale. These are the variables that—according to both the Dorfman–Steiner intuition and the extended funnel model—should matter most for advertising intensity and allocation. Inventory turnover adds a simple measure of operating efficiency that is particularly relevant for sectors like FMCG and retail, where inventory cycles are closely tied to sales.

Year dummies absorb common macro shocks (for example, the 2020 pandemic shock) and broad shifts in media markets or advertising technology. In Model 1, industry dummies capture level differences in funnel strategies across sectors. In Model 2, firm fixed effects replace industry dummies and control for any time-invariant firm traits, including industry, brand, product mix, and organizational factors.

2.4 Linking back to the theory

The theory chapter showed that, in a two-stage funnel, the optimal allocation equalizes marginal profit per dollar across upper and lower stages, and that the responsiveness of each stage to advertising depends on firm characteristics such as size, industry, and underlying effectiveness parameters. The empirical strategy here is deliberately simpler: instead of estimating the full structural model, I test whether the reduced-form outcome—the share of the budget in the upper funnel—moves in systematic ways with the observable characteristics that the model says should matter.

Model 1 answers a cross-sectional question: “At a given point in time, do firms in different industries and with different balance sheets allocate their advertising budgets differently across the funnel?” Model 2 answers a dynamic question: “When a given firm’s profitability, leverage, or size changes, does its budget shift toward or away from the upper funnel?” Together, they provide both a structural overview of where different sectors sit in the funnel space and a within-firm picture of how allocations adjust over time.

The next section turns to the regression results and interprets the key coefficients in light of these theoretical predictions.

3. Results

This section presents the main regression results and links them back to the patterns in the data and the predictions from the theory chapter. I start from the pooled industry model (Model 1), then move to the firm fixed-effects model (Model 2), and finally compare the two. Throughout, I discuss the versions with and without inventory turnover together, since the coefficients are very similar and the only difference is the loss of Comcast observations when inventory turnover is included.

For readability, the detailed regression output is reported in Appendix 4. Here I focus on the economic magnitude and sign of the key coefficients, and on how robust they are across specifications.

3.1 Model 1: Industry differences in upper-funnel allocation

3.1.1 Main coefficients

Model 1 asks how much of the variation in upper-funnel allocation can be captured by industry membership and firm characteristics in a pooled cross-section with industry and year dummies. The results are summarized in Appendix Table A.1 (without inventory turnover) and Table A.2 (with inventory turnover).

Two patterns stand out immediately. First, firm size is strongly and consistently associated with a higher upper-funnel share. In the specification without inventory turnover, the coefficient on lagged firm size is around 0.030 and highly significant. This means that increasing log assets by one unit—a rough move from the median firm to a substantially larger one—is associated with about a 3-percentage-point increase in the upper-funnel share, holding profitability, leverage, cash, industry, and year constant. The coefficient is almost identical when inventory turnover is added, which suggests that the size effect is not driven by the slightly different sample.

Second, leverage is positively related to UFS, although the magnitude is more modest. The coefficient on lagged leverage is roughly 0.009 without inventory turnover and about 0.015 with it; both estimates are statistically significant at conventional levels. Interpreted literally, this implies that moving leverage from, say, 0.2 to 0.5 (a 0.3 increase) raises UFS by about 0.3–0.5 percentage points, a small but non-negligible effect. One way to read this is that more leveraged firms may rely more on brand-building and broad awareness to sustain volumes and margins, although the effect is not as strong as the size effect.

By contrast, ROA and the cash ratio do not show robust effects in the pooled model. Their coefficients are small and statistically insignificant in both versions of Model 1. This suggests that, across firms and industries, profitability and liquidity at $t - 1$ are not strong predictors of how much of the advertising budget is allocated to the upper funnel at t , once size, leverage, and industry are taken into account.

Inventory turnover itself is essentially unrelated to UFS in this specification. The coefficient on lagged turnover is extremely close to zero and far from statistically significant, indicating that, in the cross-section, moving from slower to faster inventory cycles is not systematically associated with a higher or lower upper-funnel share once other variables are controlled for.

3.1.2 Industry effects

A key purpose of Model 1 is to quantify industry differences in UFS that were already visible in the descriptive statistics. The industry dummy coefficients in Tables A.1 and A.2 confirm the descriptive pattern: FMCG and Tech sit above the omitted base industry, while Leisure and Retail sit below.

In the specification without inventory turnover, the FMCG dummy is positive (about 0.037) and highly significant, as is the Tech dummy (around 0.039). This means that, conditional on financial controls and year dummies, FMCG and Tech firms allocate roughly 3.5–4 percentage points more of their budgets to the upper funnel than the reference industry. Leisure and Retail, by contrast, have negative and significant coefficients of about -0.050 and -0.044 respectively, indicating substantially lower UFS relative to the base industry. Healthcare and Media/Telecom do not differ significantly from the base once controls are included in this version of the model.

When inventory turnover is added (Table A.2) the broad pattern is unchanged. FMCG and Tech remain significantly above the base, Leisure and Retail remain significantly below, and the point estimates are very close to those in the simpler model. The main change is that Media/Telecom becomes significantly negative relative to the base, suggesting that once inventory turnover is controlled for and the Comcast observations drop out, media and telecom firms lean slightly more toward lower-funnel spending than the base industry.

Year dummies are mostly insignificant in both versions, with the partial exception of 2020, which is negative and marginally significant when inventory turnover is included. This is consistent with a temporary dip in upper-funnel spending during the early COVID-19 shock, but the effect is not strong enough to dominate the firm and industry patterns.

Taken together, Model 1 shows that the cross-sectional variation in UFS is dominated by two elements: firm size and persistent industry differences. Large firms and firms in FMCG and Tech allocate more of their budgets to upper-funnel activity, while smaller firms and those in Retail and Leisure allocate less, even after controlling for financial ratios and year effects.

3.2 Model 2: Within-firm dynamics

3.2.1 Main coefficients

Model 2 shifts focus from cross-sectional differences to within-firm changes over time by introducing firm fixed effects. Appendix Tables A.3 (without inventory turnover) and A.4 (with inventory turnover) report the fixed-effects estimates.

The headline result is that the strong positive association between firm size and UFS survives—and in fact remains very precise—once we look only at within-firm variation. In the fixed-effects model without inventory turnover, the coefficient on lagged firm size is about 0.024 with a t-statistic above 7. The estimate is almost identical when inventory turnover is added. This means that, when a given firm grows in size over time, its upper-funnel share tends to increase: a one-unit increase in log assets (again, a substantial change in scale) is associated with roughly a 2.4-percentage-point increase in UFS for the same firm. This is a key result: the size–UFS link is not just a cross-industry story; it also shows up as a within-firm dynamic.

Leverage is again positive, but the evidence is weaker in the fixed-effects setting. Without inventory turnover, the coefficient on lagged leverage is about 0.0088 and statistically significant at the 5 percent level. With inventory turnover included, the coefficient is similar in magnitude (about 0.0090) but only marginally significant ($p \approx 0.06$). The direction of the effect is therefore robust—when a firm becomes more leveraged, its upper-funnel share tends to rise slightly—but the precision is lower once we restrict attention to within-firm changes.

ROA and cash remain statistically insignificant in both fixed-effects models. Their coefficients are small and the t-statistics are close to zero, indicating that changes in profitability and liquidity from year to year are not systematically associated with changes in upper-funnel allocation for the same firm. This echoes the pooled results and suggests that these variables are not central drivers of funnel allocation in this sample.

Inventory turnover again plays essentially no role. In the fixed-effects model with turnover, the coefficient on lagged turnover is extremely small and insignificant. This reinforces the conclusion from Model 1: once size, leverage, and fixed effects are taken into account, there is no clear relationship between how quickly a firm turns its inventory and how much it devotes to upper-funnel advertising.

The within R^2 of Model 2 is about 0.58 in both versions, indicating that the lagged covariates and year dummies explain a sizable fraction of the within-firm variation in UFS. The estimated ρ values around 0.95 show that most of the residual variation in UFS is driven by persistent firm-specific factors, which is exactly why controlling for firm fixed effects is important.

3.2.2 Year effects and the 2020 shock

In Model 2 the year dummies absorb shocks that hit all firms simultaneously. As in Model 1, most of the year effects are small and insignificant. The 2020 dummy is negative and marginally significant in the version with inventory turnover, indicating a temporary reduction in upper-funnel allocation during the first pandemic year even after controlling for firm characteristics and fixed effects. However, the magnitude is modest (around -1.3 percentage points) and the confidence interval includes zero at the 5 percent level, so this should be interpreted as suggestive rather than definitive.

3.3 Comparing pooled and fixed-effects results

Lining up the four models highlights which relationships are purely cross-sectional and which ones survive when we look at changes within firms. Three conclusions are robust:

1. **Firm size is the central predictor of UFS.** In every specification—pooled with industry dummies, and fixed effects with firm dummies, with and without inventory turnover—the coefficient on lagged firm size is positive, sizable, and highly significant. Cross-sectionally, larger firms allocate more to the upper funnel; within firms, increases in size go hand in hand with increases in upper-funnel share.
2. **Leverage matters, but modestly.** Leverage is consistently positive, with significant or marginally significant coefficients across models. The magnitudes are smaller than for size, but the direction is stable: more debt is associated with slightly higher upper-funnel allocation.
3. **ROA, cash, and inventory turnover are not key drivers.** Profitability, liquidity, and inventory turnover do not show robust effects in either the pooled or fixed-effects models. Their coefficients bounce around zero and are never statistically strong, suggesting that they are at best second-order determinants of funnel allocation in this particular sample.

The industry dummies in Model 1 add another layer: even after we control for firm-level covariates, some industries systematically sit above or below the baseline in terms of UFS. FMCG and Tech are reliably more upper-funnel-intensive, while Retail and Leisure are reliably less so. Model 2, by contrast, absorbs these level differences into the firm fixed effects and focuses on dynamics. The fact that the size and leverage results survive this much stricter test supports the interpretation that they capture genuine within-firm relationships rather than pure cross-firm sorting.

3.4 Interpretation

The empirical results line up well with the theoretical framework. The theory predicts that firms with stronger underlying effectiveness in upper-funnel stages, larger installed bases, or broader markets will find it optimal to tilt more of their budgets toward awareness and consideration. In

practice, these advantages tend to come with scale: large FMCG and tech firms have massive brands and reach, and they show up exactly where the model says they should—in the high-UFS part of the distribution, even after controlling for other factors.

The positive role of leverage is also plausible. More leveraged firms face higher pressure to maintain cash flows and may view broad brand investment as a way to stabilize demand and pricing power over time, even if this requires sacrificing some short-run performance marketing. The effect is not huge, but it is consistent across specifications.

The absence of strong effects for ROA, cash, and inventory turnover is equally informative. Within this group of large, established firms, upper-funnel allocation seems to be a strategic stance that is tied more to scale and industry positioning than to year-to-year movements in profitability, liquidity, or operational efficiency. Put differently, mature firms do not appear to swing their funnel allocations dramatically in response to short-run financial fluctuations; instead, they maintain relatively stable upper-funnel shares that move gradually with size and capital structure.

3.5 Limitations

The regression results should be read with a few important limitations in mind. First, the dataset is deliberately narrow: it covers 18 large, listed firms across seven industries, all of which are already well established and operate in consumer-facing markets. The youngest firm in the sample is still a mature advertiser, so the analysis cannot speak to early-stage or high-growth firms whose funnel strategies may evolve much more dramatically over time. Relatedly, the average firm in the sample is very large in terms of assets and advertising spend, which means the results are best interpreted as describing big-brand allocation behavior, not the full universe of firms.

Second, the panel is modest in size—270 firm-years overall, and 254 when inventory turnover is included—and was constructed manually from annual reports. Building the dataset observation by observation imposes a practical time constraint: expanding the sample to more firms, more years, or additional countries would quickly become infeasible at this level of manual detail. As a result, some choices were made for tractability rather than ideal coverage, such as focusing on

a single Comcast-style inventory reporting issue rather than fully harmonizing every possible accounting difference. The relatively small number of firms also limits statistical power, especially for detecting more subtle effects of ROA, cash, or inventory turnover.

Third, the empirical design relies on an imputed upper-funnel share rather than a directly observed allocation. The mapping from log advertising spend to UFS is grounded in the theoretical model and calibrated to produce realistic shares, but it is still a modeling choice rather than a disclosed firm metric. If the true relationship between size and funnel allocation differs from the assumed logit form, some of the estimated coefficients may partly reflect misspecification of that mapping. In this sense, the regressions test how accounting variables relate to a theory-consistent measure of UFS, not to firms' actual internal budget splits.

Fourth, several potentially relevant variables are omitted or only partially observed. R&D spending, for example, had to be dropped because missing values were too frequent, even though it likely correlates with long-term brand building and innovation. The regressions also abstract from richer aspects of digital maturity, media mix, or competitive environment that are difficult to measure from public reports but could shape funnel strategies. To the extent that these omitted factors move together with size or leverage, some of the estimated effects may capture broader "firm type" differences rather than pure balance-sheet mechanisms.

Finally, the analysis uses annual data and one-year lags, which is appropriate for budget planning but too coarse to capture within-year dynamics, campaign cycles, or shorter-run responses to shocks. The fixed-effects models control for time-invariant firm traits, but they cannot fully address potential reverse causality—for instance, if persistent funnel strategies influence future profitability or leverage rather than the other way around.

Taken together, these limitations mean the results should be interpreted as stylized patterns for a specific set of large, mature firms, rather than universal laws for all advertisers. They provide useful evidence on how firm characteristics correlate with theory-consistent funnel allocations, but there is room for future work with richer data and broader samples to sharpen and extend these findings.

4. Appendix

Model 1.1 — Industry Differences (No Firm Fixed Effects) [without Inventory Turnover]

reg UFS ROA_Lag Cash_Lag Leverage_Lag FirmSize i.Industry_N i.Year

Table A1:

Results						
Source	SS	df	MS	Number of obs	=	269
Model	2.13976571	24	.089156905	F(24, 244)	=	135.19
Residual	.160917284	244	.000659497	Prob > F	=	0.0000
				R-squared	=	0.9301
				Adj R-squared	=	0.9232
Total	2.300683	268	.008584638	Root MSE	=	.02568
UFS	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ROA_Lag	.031807	.0222761	1.43	0.155	-.0120709	.0756849
Cash_Lag	-.0175397	.0253544	-0.69	0.490	-.0674811	.0324017
Leverage_Lag	.0089527	.0035212	2.54	0.012	.002017	.0158885
FirmSize	.0301171	.0020393	14.77	0.000	.0261002	.0341339
Industry_N						
FMCG	.0365479	.006303	5.80	0.000	.0241326	.0489631
Healthcare	-.0098088	.0069793	-1.41	0.161	-.0235561	.0039386
Leisure	-.0497586	.0127774	-3.89	0.000	-.0749266	-.0245906
Media/Telecom	-.000552	.0069083	-0.08	0.936	-.0141594	.0130554
Retail	-.0440281	.0107833	-4.08	0.000	-.0652683	-.0227878
Tech	.0394199	.0093262	4.23	0.000	.0210497	.0577901
Year						
2011	.0009585	.0088651	0.11	0.914	-.0165034	.0184203
2012	.0032608	.008868	0.37	0.713	-.0142068	.0207285
2013	.0016314	.008847	0.18	0.854	-.0157948	.0190576
2014	.0049819	.0089345	0.56	0.578	-.0126167	.0225806
2015	.0055421	.0087776	0.63	0.528	-.0117475	.0228317
2016	.003202	.0088625	0.36	0.718	-.0142548	.0206588
2017	.0018628	.0088743	0.21	0.834	-.0156172	.0193427
2018	.0016922	.008964	0.19	0.850	-.0159645	.0193489
2019	-.000123	.0089474	-0.01	0.989	-.017747	.0175009
2020	-.0121266	.0089175	-1.36	0.175	-.0296916	.0054384
2021	-.0041893	.0088596	-0.47	0.637	-.0216403	.0132618
2022	-.0038521	.0089644	-0.43	0.668	-.0215096	.0138053
2023	-.0063997	.0089831	-0.71	0.477	-.0240939	.0112946
2024	-.007806	.0089661	-0.87	0.385	-.025467	.0098549

Model 1.2 — Industry Differences (No Firm Fixed Effects) [with Inventory Turnover]

reg UFS ROA_Lag Cash_Lag Leverage_Lag FirmSize InventoryTurnover_Lag i.Industry_N i.Year

Table A2:

Results						
Source	SS	df	MS	Number of obs = 254		
Model	2.06162411	25	.082464964	F(25, 228)	=	130.28
Residual	.144324168	228	.000633001	Prob > F	=	0.0000
				R-squared	=	0.9346
				Adj R-squared	=	0.9274
Total	2.20594828	253	.008719163	Root MSE	=	.02516
UFS	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ROA_Lag	.0370922	.0237123	1.56	0.119	-.0096311	.0838154
Cash_Lag	-.0069735	.0256607	-0.27	0.786	-.057536	.043589
Leverage_Lag	.0147154	.0037137	3.96	0.000	.0073978	.022033
FirmSize_Lag	.0304506	.0019787	15.39	0.000	.0265517	.0343495
InventoryTurnover_Lag	-.0000147	.0002143	-0.07	0.945	-.000437	.0004076
Industry_N						
FMCG	.0372628	.0065212	5.71	0.000	.0244132	.0501124
Healthcare	-.0093572	.0073003	-1.28	0.201	-.0237418	.0050275
Leisure	-.0479638	.0125045	-3.84	0.000	-.072603	-.0233247
Media/Telecom	-.0186278	.0091386	-2.04	0.043	-.0366346	-.000621
Retail	-.0386427	.0110886	-3.48	0.001	-.0604919	-.0167936
Tech	.0456649	.0091319	5.00	0.000	.0276713	.0636585
Year						
2011	-.0011625	.0090167	-0.13	0.898	-.0189292	.0166041
2012	-.00149	.0091246	-0.16	0.870	-.0194694	.0164894
2013	.0006089	.009114	0.07	0.947	-.0173495	.0185672
2014	-.0012542	.0091609	-0.14	0.891	-.0193051	.0167967
2015	-.0006618	.0090509	-0.07	0.942	-.0184958	.0171723
2016	-.0026865	.0090259	-0.30	0.766	-.0204713	.0150982
2017	-.0033392	.009064	-0.37	0.713	-.0211991	.0145208
2018	-.0042381	.0091401	-0.46	0.643	-.0222481	.0137718
2019	-.0029042	.0091166	-0.32	0.750	-.0208678	.0150593
2020	-.0175626	.0090841	-1.93	0.054	-.0354622	.0003369
2021	-.0113316	.0090292	-1.25	0.211	-.0291229	.0064598
2022	-.0119178	.0091068	-1.31	0.192	-.0298621	.0060265
2023	-.0140791	.0091255	-1.54	0.124	-.0320602	.003902
2024	-.0133522	.0090871	-1.47	0.143	-.0312577	.0045532

Model 2.1 — Within-Firm Dynamics (Firm Fixed Effects) [without Inventory Turnover]

xtreg UFS ROA_Lag Cash_Lag Leverage_Lag FirmSize_Lag i.Year, fe robust

Table A3:

Fixed-effects (within) regression				Number of obs	=	269
Group variable: Firm_N				Number of groups	=	18
R-squared:				Obs per group:		
Within = 0.5813				min	=	14
Between = 0.8576				avg	=	14.9
Overall = 0.8422				max	=	15
corr(u_i, Xb) = 0.6727				F(17, 17)	=	.
				Prob > F	=	.
(Std. err. adjusted for 18 clusters in Firm_N)						
UFS	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ROA_Lag	.0004635	.0136546	0.03	0.973	-.0283452	.0292723
Cash_Lag	-.0542029	.0472895	-1.15	0.268	-.153975	.0455693
Leverage_Lag	.0087711	.0040675	2.16	0.046	.0001894	.0173528
FirmSize_Lag	.0240707	.0033753	7.13	0.000	.0169495	.0311919
Year						
2011	.001228	.0031133	0.39	0.698	-.0053405	.0077966
2012	.0009462	.0028737	0.33	0.746	-.0051169	.0070092
2013	.0033063	.0025537	1.29	0.213	-.0020814	.008694
2014	.0024533	.0026972	0.91	0.376	-.0032372	.0081439
2015	.0038489	.0032448	1.19	0.252	-.002997	.0106948
2016	.0034995	.0034996	1.00	0.331	-.003884	.010883
2017	.0031767	.00422	0.75	0.462	-.0057269	.0120802
2018	.0032846	.0051963	0.63	0.536	-.0076786	.0142478
2019	.0035123	.0053899	0.65	0.523	-.0078593	.0148839
2020	-.0104885	.0065609	-1.60	0.128	-.0243307	.0033538
2021	-.0044683	.0051425	-0.87	0.397	-.0153181	.0063814
2022	-.0029662	.0063484	-0.47	0.646	-.0163602	.0104278
2023	-.0054531	.0083713	-0.65	0.523	-.0231149	.0122087
2024	-.0049855	.0081081	-0.61	0.547	-.0220921	.012121
_cons	.3073179	.0327355	9.39	0.000	.2382521	.3763836
sigma_u	.04889549					
sigma_e	.01170097					
rho	.94583466	(fraction of variance due to u_i)				

Model 2.2 — Within-Firm Dynamics (Firm Fixed Effects) [with Inventory Turnover]

xtreg UFS ROA_Lag Cash_Lag Leverage_Lag FirmSize_Lag InventoryTurnover_Lag i.Year, fe robust

Table A4:

Fixed-effects (within) regression		Number of obs	=	254			
Group variable: Firm_N		Number of groups	=	17			
R-squared:		Obs per group:					
Within	= 0.5813	min	=	14			
Between	= 0.8487	avg	=	14.9			
Overall	= 0.8337	max	=	15			
corr(u_i, Xb) = 0.6567		F(16, 16)	=	.			
		Prob > F	=	.			
(Std. err. adjusted for 17 clusters in Firm_N)							
UFS	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]		
ROA_Lag	.0001114	.0137264	0.01	0.994	-.0289874	.0292102	
Cash_Lag	-.0578542	.0482894	-1.20	0.248	-.1602233	.0445148	
Leverage_Lag	.0089742	.0045041	1.99	0.064	-.0005741	.0185225	
FirmSize_Lag	.024086	.0033199	7.25	0.000	.017048	.031124	
InventoryTurnover_Lag	.000016	.000141	0.11	0.911	-.0002829	.0003148	
Year							
2011	-.0001472	.0031605	-0.05	0.963	-.0068472	.0065529	
2012	-.0004123	.0029987	-0.14	0.892	-.0067692	.0059446	
2013	.0020678	.0028976	0.71	0.486	-.0040748	.0082105	
2014	.0010848	.002572	0.42	0.679	-.0043675	.0065371	
2015	.0022193	.003198	0.69	0.498	-.0045603	.0089988	
2016	.0018279	.0034446	0.53	0.603	-.0054744	.0091302	
2017	.0015755	.0042439	0.37	0.715	-.0074212	.0105722	
2018	.0015223	.0052009	0.29	0.774	-.0095031	.0125478	
2019	.0019288	.0054989	0.35	0.730	-.0097285	.013586	
2020	-.0126592	.0065833	-1.92	0.072	-.0266151	.0012967	
2021	-.0065061	.0049161	-1.32	0.204	-.0169278	.0039156	
2022	-.0050783	.0062387	-0.81	0.428	-.0183036	.0081471	
2023	-.0077107	.0085158	-0.91	0.379	-.0257634	.010342	
2024	-.0072053	.0082882	-0.87	0.398	-.0247756	.010365	
_cons	.3070638	.0320339	9.59	0.000	.239155	.3749726	
sigma_u	.04983917						
sigma_e	.01190405						
rho	.94602995	(fraction of variance due to u_i)					

